# Multi-seed results

32-bit phase data, LoRA rank, loss and seed

Results of 180 LoRA training runs: 3 models (EdgeSAM, MobileSAM, MobileNet-UNet), 5 ranks (2, 4, 8, 16, 32), 4 loss settings and 3 seeds. Source: `results.json`, generated 2026-10-24T23:59:34.450309+00:00. Class names follow the manuscript order: mask value 1 = discocyte, mask value 2 = spherocyte, mask value 3 = echinocyte, mask value 4 = stomatocyte.

# How to read the numbers

**Dice** overlap between predicted and annotated cells, averaged over the four cell classes. 1 is perfect, higher is better.

**AJI, BF1** AJI: how well single cells are separated. BF1: how well the cell outlines match. Both 0 to 1, higher is better.

**IPE** error of the integrated phase (the quantity that gives the dry mass), in percent. Lower is better.

**Lost class** a run in which at least one cell class has Dice below 0.05, so that class is practically not found. Such runs are kept in every mean. A small superscript number next to a value is the number of runs, out of 3 seeds, that lost a class.

**Seeds** each setting was trained 3 times with different random seeds (42, 43, 44). ± is the standard deviation over these seeds. With three seeds the numbers describe what happened; they are not significance tests.

**Setting** 32-bit phase maps in radians, last-epoch model, 66 validation images. The rank is the LoRA rank (2 to 32). Losses: Dice only, +PMC, +PMC+BGA, Full (+IPP).

# Key points

- **EdgeSAM.** With the full objective, the mean Dice over seeds is highest at rank 32 (0.738) and lowest at rank 2 (0.625). 3 of 15 full-objective runs lost a class (at rank 2, 8, 16); over all four losses 11 of 60 runs did. Classes lost most often: spherocyte (11). The full objective has a lower global IPE than Dice only in 15 of 15 seed-by-seed comparisons (same rank and seed).
- **MobileSAM.** With the full objective, the mean Dice over seeds is highest at rank 16 (0.813) and lowest at rank 32 (0.687). 1 of 15 full-objective runs lost a class (at rank 32); over all four losses 6 of 60 runs did. Classes lost most often: spherocyte (6), stomatocyte (1). The full objective has a lower global IPE than Dice only in 14 of 15 seed-by-seed comparisons (same rank and seed).
- **MobileNet-UNet.** With the full objective, the mean Dice over seeds is highest at rank 32 (0.476) and lowest at rank 4 (0.406). 15 of 15 full-objective runs lost a class (at rank 2, 4, 8, 16, 32); over all four losses 56 of 60 runs did. Classes lost most often: spherocyte (54), stomatocyte (43), echinocyte (10). The full objective has a lower global IPE than Dice only in 13 of 15 seed-by-seed comparisons (same rank and seed).

# Full objective at each rank

| | EdgeSAM | | | MobileSAM | | | MobileNet-UNet | | |
|---|---|---|---|---|---|---|---|---|---|
| Rank | Dice | IPE (%) | Lost | Dice | IPE (%) | Lost | Dice | IPE (%) | Lost |
| 2 | 0.625 ± 0.108 | 4.45 ± 2.71 | 1/3 | 0.808 ± 0.003 | 1.30 ± 0.09 | 0/3 | 0.410 ± 0.004 | 2.61 ± 0.23 | 3/3 |
| 4 | 0.703 ± 0.009 | 3.48 ± 0.91 | 0/3 | 0.811 ± 0.004 | 1.34 ± 0.07 | 0/3 | 0.406 ± 0.002 | 2.50 ± 0.19 | 3/3 |
| 8 | 0.647 ± 0.114 | 3.81 ± 1.82 | 1/3 | 0.810 ± 0.006 | 1.32 ± 0.09 | 0/3 | 0.412 ± 0.004 | 2.60 ± 0.07 | 3/3 |
| 16 | 0.665 ± 0.114 | 4.66 ± 3.53 | 1/3 | 0.813 ± 0.002 | 1.29 ± 0.18 | 0/3 | 0.414 ± 0.002 | 3.09 ± 0.65 | 3/3 |
| 32 | 0.738 ± 0.006 | 3.02 ± 0.84 | 0/3 | 0.687 ± 0.234 | 1.76 ± 0.53 | 1/3 | 0.476 ± 0.108 | 5.83 ± 5.75 | 3/3 |

**Table 1:** Full objective at each LoRA rank (mean ± SD over 3 seeds). "Lost" = runs that lost a class.

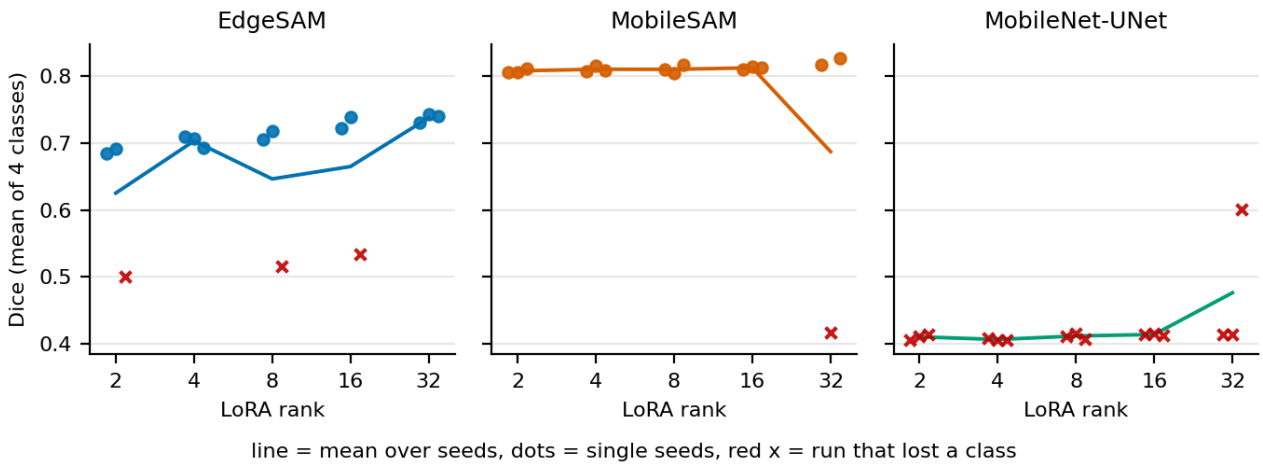



**Figure 1:** Dice by LoRA rank, Full (+IPP) objective. Higher is better.

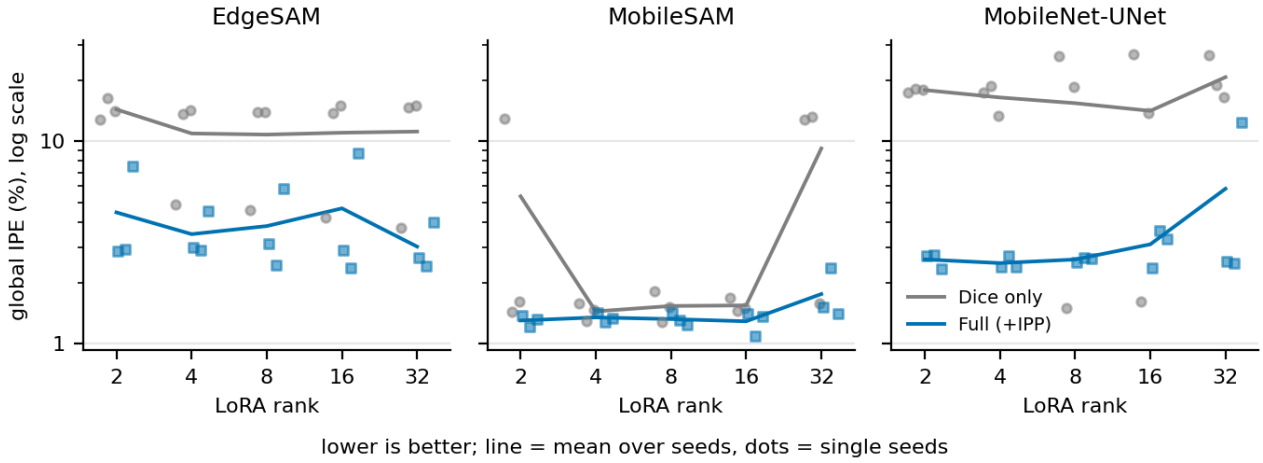



**Figure 2:** Global IPE by LoRA rank, Dice only versus Full (+IPP) objective. Lower is better.

## Effect of the loss, at every rank

| Model | Loss | r = 2 | r = 4 | r = 8 | r = 16 | r = 32 |
|---|---|---|---|---|---|---|
| EdgeSAM | Dice only | 0.576 | 0.610[1] | 0.624[1] | 0.621[2] | 0.619[2] |
| | +PMC | 0.710 | 0.727 | 0.672[1] | 0.741 | 0.680[1] |
| | +PMC+BGA | 0.717 | 0.723 | 0.735 | 0.742 | 0.746 |
| | Full (+IPP) | 0.625[1] | 0.703 | 0.647[1] | 0.665[1] | 0.738 |
| MobileSAM | Dice only | 0.741[1] | 0.811 | 0.811 | 0.813 | 0.674[2] |
| | +PMC | 0.805 | 0.806 | 0.813 | 0.813 | 0.817 |
| | +PMC+BGA | 0.740[1] | 0.741[1] | 0.813 | 0.817 | 0.818 |
| | Full (+IPP) | 0.808 | 0.811 | 0.810 | 0.813 | 0.687[1] |
| MobileNet-UNet | Dice only | 0.416[3] | 0.482[3] | 0.485[2] | 0.550[2] | 0.359[3] |
| | +PMC | 0.413[3] | 0.417[3] | 0.414[3] | 0.413[3] | 0.486[3] |
| | +PMC+BGA | 0.549[3] | 0.490[3] | 0.544[2] | 0.612[2] | 0.418[3] |
| | Full (+IPP) | 0.410[3] | 0.406[3] | 0.412[3] | 0.414[3] | 0.476[3] |

**Table 2:** Dice by loss and rank (mean over 3 seeds). Superscript = runs that lost a class. Higher is better.

| Model | Loss | r = 2 | r = 4 | r = 8 | r = 16 | r = 32 |
|---|---|---|---|---|---|---|
| EdgeSAM | Dice only | 14.42 | 10.96[1] | 10.82[1] | 11.05[2] | 11.20[2] |
| | +PMC | 5.26 | 4.68 | 7.97[1] | 3.71 | 7.45[1] |
| | +PMC+BGA | 4.88 | 4.98 | 4.25 | 3.43 | 3.64 |
| | Full (+IPP) | 4.45[1] | 3.48 | 3.81[1] | 4.66[1] | 3.02 |
| MobileSAM | Dice only | 5.35[1] | 1.44 | 1.53 | 1.54 | 9.23[2] |
| | +PMC | 1.55 | 1.50 | 1.60 | 1.58 | 1.58 |
| | +PMC+BGA | 5.32[1] | 5.23[1] | 1.48 | 1.55 | 1.46 |
| | Full (+IPP) | 1.30 | 1.34 | 1.32 | 1.29 | 1.76[1] |
| MobileNet-UNet | Dice only | 17.98[3] | 16.52[3] | 15.48[2] | 14.18[2] | 20.83[3] |
| | +PMC | 19.88[3] | 19.39[3] | 19.79[3] | 20.13[3] | 15.96[3] |
| | +PMC+BGA | 12.40[3] | 14.70[3] | 13.96[2] | 11.50[2] | 19.54[3] |
| | Full (+IPP) | 2.61[3] | 2.50[3] | 2.60[3] | 3.09[3] | 5.83[3] |

**Table 3:** Global IPE in percent by loss and rank (mean over 3 seeds). Superscript = runs that lost a class. Lower is better.

# Dice of each cell class

| Model | Rank | Disco | Sphero | Echi | Stoma |
|---|---|---|---|---|---|
| EdgeSAM | 2 | 0.882 ± 0.008 | 0.497 ± 0.431 | 0.616 ± 0.010 | 0.506 ± 0.013 |
| | 4 | 0.891 ± 0.002 | 0.766 ± 0.013 | 0.612 ± 0.007 | 0.544 ± 0.025 |
| | 8 | 0.882 ± 0.012 | 0.513 ± 0.444 | 0.628 ± 0.006 | 0.563 ± 0.019 |
| | 16 | 0.890 ± 0.007 | 0.521 ± 0.451 | 0.659 ± 0.015 | 0.591 ± 0.027 |
| | 32 | 0.895 ± 0.001 | 0.789 ± 0.005 | 0.655 ± 0.010 | 0.614 ± 0.019 |
| MobileSAM | 2 | 0.924 ± 0.000 | 0.848 ± 0.002 | 0.767 ± 0.001 | 0.694 ± 0.013 |
| | 4 | 0.923 ± 0.002 | 0.855 ± 0.005 | 0.773 ± 0.003 | 0.691 ± 0.021 |
| | 8 | 0.923 ± 0.003 | 0.851 ± 0.004 | 0.760 ± 0.006 | 0.708 ± 0.020 |
| | 16 | 0.923 ± 0.003 | 0.855 ± 0.002 | 0.777 ± 0.007 | 0.696 ± 0.009 |
| | 32 | 0.913 ± 0.020 | 0.574 ± 0.497 | 0.777 ± 0.004 | 0.485 ± 0.420 |
| MobileNet-UNet | 2 | 0.889 ± 0.001 | 0.000 ± 0.000 | 0.753 ± 0.016 | 0.000 ± 0.000 |
| | 4 | 0.889 ± 0.001 | 0.000 ± 0.000 | 0.737 ± 0.005 | 0.000 ± 0.000 |
| | 8 | 0.891 ± 0.002 | 0.000 ± 0.000 | 0.756 ± 0.015 | 0.000 ± 0.000 |
| | 16 | 0.892 ± 0.001 | 0.000 ± 0.000 | 0.764 ± 0.006 | 0.000 ± 0.000 |
| | 32 | 0.903 ± 0.022 | 0.000 ± 0.000 | 0.767 ± 0.003 | 0.236 ± 0.408 |

**Table 4:** Dice of each cell class, full objective (mean ± SD over seeds). A class with Dice near 0 is not found by that model. Class order: discocyte, spherocyte, echinocyte, stomatocyte.

The complete tables (every seed, every metric, every run, paired differences) follow in the appendix.

## Scope and conventions

- Source: `results.json` (generated 2026-10-24T23:59:34.450309+00:00). 180 LoRA runs: 3 architectures × 5 ranks × 4 losses × 3 seeds (42, 43, 44).
- Checkpoint: last epoch. Validation images: 66. Input: 32-bit phase maps in radians (no 8-bit conversion).
- Losses: Dice only, +PMC, +PMC+BGA, Full (+IPP). Ranks: 2, 4, 8, 16, 32.
- A run *lost a class* when the Dice of any class is below 0.05. Such runs are counted in every mean, none is excluded.
- Mean ± SD over seeds (sample SD, $n$ as stated). IPE is the global integrated phase error in percent. Per-class columns follow the mask-value order: discocyte, spherocyte, echinocyte, stomatocyte (mask values 1 = discocyte, 2 = spherocyte, 3 = echinocyte, 4 = stomatocyte).
- Paired differences are computed seed by seed (same seed in both runs), with a 95% $t$-interval; with three seeds the interval is wide and is given for description only, not as a significance test.

# A Rank sweep, full objective

**Table 5:** EdgeSAM: LoRA rank sweep, Full (+IPP) objective. † = run lost a class (Dice < 0.05).

| Rank | Params (M) | Dice | AJI | BF1 | IPE (%) | Runs that lost a class | Dice s42 | Dice s43 | Dice s44 | IPE s42 | IPE s43 | IPE s44 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 0.30 | 0.625 ± 0.108 | 0.249 ± 0.017 | 0.598 ± 0.020 | 4.45 ± 2.71 | 1/3 (seed 44) | 0.685 | 0.691 | 0.501† | 2.86 | 2.93 | 7.58† |
| 4 | 0.38 | 0.703 ± 0.009 | 0.252 ± 0.021 | 0.615 ± 0.012 | 3.48 ± 0.91 | 0/3 | 0.710 | 0.707 | 0.693 | 2.99 | 2.91 | 4.53 |
| 8 | 0.53 | 0.647 ± 0.114 | 0.249 ± 0.043 | 0.608 ± 0.009 | 3.81 ± 1.82 | 1/3 (seed 44) | 0.706 | 0.719 | 0.515† | 3.13 | 2.43 | 5.87† |
| 16 | 0.85 | 0.665 ± 0.114 | 0.277 ± 0.022 | 0.623 ± 0.024 | 4.66 ± 3.53 | 1/3 (seed 44) | 0.722 | 0.740 | 0.533† | 2.90 | 2.37 | 8.73† |
| 32 | 1.47 | 0.738 ± 0.006 | 0.335 ± 0.020 | 0.642 ± 0.005 | 3.02 ± 0.84 | 0/3 | 0.731 | 0.743 | 0.740 | 2.66 | 2.41 | 3.98 |

**Table 6:** MobileSAM: LoRA rank sweep, Full (+IPP) objective. † = run lost a class (Dice < 0.05).

| Rank | Params (M) | Dice | AJI | BF1 | IPE (%) | Runs that lost a class | Dice s42 | Dice s43 | Dice s44 | IPE s42 | IPE s43 | IPE s44 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 0.63 | 0.808 ± 0.003 | 0.264 ± 0.013 | 0.801 ± 0.006 | 1.30 ± 0.09 | 0/3 | 0.806 | 0.807 | 0.812 | 1.38 | 1.21 | 1.31 |
| 4 | 0.64 | 0.811 ± 0.004 | 0.268 ± 0.037 | 0.803 ± 0.002 | 1.34 ± 0.07 | 0/3 | 0.807 | 0.815 | 0.809 | 1.42 | 1.28 | 1.33 |
| 8 | 0.67 | 0.810 ± 0.006 | 0.262 ± 0.017 | 0.802 ± 0.011 | 1.32 ± 0.09 | 0/3 | 0.810 | 0.805 | 0.817 | 1.42 | 1.30 | 1.24 |
| 16 | 0.73 | 0.813 ± 0.002 | 0.279 ± 0.054 | 0.801 ± 0.013 | 1.29 ± 0.18 | 0/3 | 0.811 | 0.814 | 0.813 | 1.41 | 1.08 | 1.37 |
| 32 | 0.85 | 0.687 ± 0.234 | 0.283 ± 0.020 | 0.794 ± 0.023 | 1.76 ± 0.53 | 1/3 (seed 43) | 0.818 | 0.417† | 0.827 | 1.51 | 2.36† | 1.40 |

**Table 7:** MobileNet-UNet: LoRA rank sweep, Full (+IPP) objective. † = run lost a class (Dice < 0.05).

| Rank | Params (M) | Dice | AJI | BF1 | IPE (%) | Runs that lost a class | Dice s42 | Dice s43 | Dice s44 | IPE s42 | IPE s43 | IPE s44 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 5.47 | 0.410 ± 0.004 | 0.534 ± 0.029 | 0.816 ± 0.006 | 2.61 ± 0.23 | 3/3 (seed 42, 43, 44) | 0.406† | 0.411† | 0.414† | 2.73† | 2.75† | 2.34† |

| Rank | Params (M) | Dice | AJI | BF1 | IPE (%) | Runs that lost a class | Dice s42 | Dice s43 | Dice s44 | IPE s42 | IPE s43 | IPE s44 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 5.51 | 0.406 ± 0.002 | 0.560 ± 0.035 | 0.805 ± 0.006 | 2.50 ± 0.19 | 3/3 (seed 42, 43, 44) | 0.408† | 0.406† | 0.405† | 2.39† | 2.72† | 2.39† |
| 8 | 5.58 | 0.412 ± 0.004 | 0.551 ± 0.053 | 0.814 ± 0.003 | 2.60 ± 0.07 | 3/3 (seed 42, 43, 44) | 0.412† | 0.416† | 0.407† | 2.52† | 2.66† | 2.63† |
| 16 | 5.73 | 0.414 ± 0.002 | 0.505 ± 0.012 | 0.805 ± 0.009 | 3.09 ± 0.65 | 3/3 (seed 42, 43, 44) | 0.413† | 0.416† | 0.413† | 2.37† | 3.63† | 3.28† |
| 32 | 6.04 | 0.476 ± 0.108 | 0.541 ± 0.049 | 0.824 ± 0.014 | 5.83 ± 5.75 | 3/3 (seed 42, 43, 44) | 0.414† | 0.414† | 0.601† | 2.54† | 2.49† | 12.47† |

# B Loss ablation at every rank

**Table 8:** EdgeSAM: all losses at all ranks (mean ± SD over seeds).

| Rank | Loss | Params (M) | Dice | AJI | BF1 | IPE (%) | Runs that lost a class |
|---|---|---|---|---|---|---|---|
| 2 | Dice only | 0.30 | 0.576 ± 0.013 | 0.311 ± 0.034 | 0.535 ± 0.063 | 14.42 ± 1.81 | 0/3 |
| 2 | +PMC | 0.30 | 0.710 ± 0.005 | 0.374 ± 0.061 | 0.606 ± 0.027 | 5.26 ± 1.37 | 0/3 |
| 2 | +PMC+BGA | 0.30 | 0.717 ± 0.006 | 0.385 ± 0.055 | 0.610 ± 0.021 | 4.88 ± 1.59 | 0/3 |
| 2 | Full (+IPP) | 0.30 | 0.625 ± 0.108 | 0.249 ± 0.017 | 0.598 ± 0.020 | 4.45 ± 2.71 | 1/3 (seed 44) |
| 4 | Dice only | 0.38 | 0.610 ± 0.085 | 0.321 ± 0.057 | 0.476 ± 0.210 | 10.96 ± 5.30 | 1/3 (seed 43) |
| 4 | +PMC | 0.38 | 0.727 ± 0.009 | 0.363 ± 0.032 | 0.619 ± 0.021 | 4.68 ± 1.20 | 0/3 |
| 4 | +PMC+BGA | 0.38 | 0.723 ± 0.011 | 0.392 ± 0.039 | 0.614 ± 0.025 | 4.98 ± 1.57 | 0/3 |
| 4 | Full (+IPP) | 0.38 | 0.703 ± 0.009 | 0.252 ± 0.021 | 0.615 ± 0.012 | 3.48 ± 0.91 | 0/3 |
| 8 | Dice only | 0.53 | 0.624 ± 0.095 | 0.334 ± 0.052 | 0.496 ± 0.209 | 10.82 ± 5.39 | 1/3 (seed 43) |
| 8 | +PMC | 0.53 | 0.672 ± 0.109 | 0.386 ± 0.033 | 0.621 ± 0.031 | 7.97 ± 6.80 | 1/3 (seed 44) |
| 8 | +PMC+BGA | 0.53 | 0.735 ± 0.010 | 0.390 ± 0.046 | 0.628 ± 0.013 | 4.25 ± 1.07 | 0/3 |
| 8 | Full (+IPP) | 0.53 | 0.647 ± 0.114 | 0.249 ± 0.043 | 0.608 ± 0.009 | 3.81 ± 1.82 | 1/3 (seed 44) |
| 16 | Dice only | 0.85 | 0.621 ± 0.103 | 0.315 ± 0.047 | 0.507 ± 0.221 | 11.05 ± 5.98 | 2/3 (seed 43, 44) |
| 16 | +PMC | 0.85 | 0.741 ± 0.010 | 0.398 ± 0.077 | 0.643 ± 0.009 | 3.71 ± 0.71 | 0/3 |
| 16 | +PMC+BGA | 0.85 | 0.742 ± 0.006 | 0.339 ± 0.024 | 0.644 ± 0.009 | 3.43 ± 0.81 | 0/3 |
| 16 | Full (+IPP) | 0.85 | 0.665 ± 0.114 | 0.277 ± 0.022 | 0.623 ± 0.024 | 4.66 ± 3.53 | 1/3 (seed 44) |
| 32 | Dice only | 1.47 | 0.619 ± 0.109 | 0.329 ± 0.015 | 0.519 ± 0.212 | 11.20 ± 6.46 | 2/3 (seed 43, 44) |
| 32 | +PMC | 1.47 | 0.680 ± 0.113 | 0.388 ± 0.059 | 0.638 ± 0.027 | 7.45 ± 7.17 | 1/3 (seed 44) |
| 32 | +PMC+BGA | 1.47 | 0.746 ± 0.006 | 0.406 ± 0.034 | 0.643 ± 0.011 | 3.64 ± 0.82 | 0/3 |
| 32 | Full (+IPP) | 1.47 | 0.738 ± 0.006 | 0.335 ± 0.020 | 0.642 ± 0.005 | 3.02 ± 0.84 | 0/3 |

**Table 9:** MobileSAM: all losses at all ranks (mean ± SD over seeds).

| Rank | Loss | Params (M) | Dice | AJI | BF1 | IPE (%) | Runs that lost a class |
|---|---|---|---|---|---|---|---|
| 2 | Dice only | 0.63 | 0.741 ± 0.123 | 0.219 ± 0.036 | 0.779 ± 0.035 | 5.35 ± 6.63 | 1/3 (seed 42) |
| 2 | +PMC | 0.63 | 0.805 ± 0.007 | 0.228 ± 0.026 | 0.800 ± 0.003 | 1.55 ± 0.12 | 0/3 |
| 2 | +PMC+BGA | 0.63 | 0.740 ± 0.125 | 0.237 ± 0.026 | 0.776 ± 0.036 | 5.32 ± 6.59 | 1/3 (seed 42) |
| 2 | Full (+IPP) | 0.63 | 0.808 ± 0.003 | 0.264 ± 0.013 | 0.801 ± 0.006 | 1.30 ± 0.09 | 0/3 |
| 4 | Dice only | 0.64 | 0.811 ± 0.002 | 0.268 ± 0.025 | 0.801 ± 0.007 | 1.44 ± 0.15 | 0/3 |
| 4 | +PMC | 0.64 | 0.806 ± 0.006 | 0.236 ± 0.025 | 0.799 ± 0.005 | 1.50 ± 0.13 | 0/3 |

| Rank | Loss | Params (M) | Dice | AJI | BF1 | IPE (%) | Runs that lost a class |
|---|---|---|---|---|---|---|---|
| 4 | +PMC+BGA | 0.64 | 0.741 ± 0.122 | 0.247 ± 0.037 | 0.783 ± 0.036 | 5.23 ± 6.58 | 1/3 (seed 42) |
| 4 | Full (+IPP) | 0.64 | 0.811 ± 0.004 | 0.268 ± 0.037 | 0.803 ± 0.002 | 1.34 ± 0.07 | 0/3 |
| 8 | Dice only | 0.67 | 0.811 ± 0.005 | 0.248 ± 0.022 | 0.808 ± 0.012 | 1.53 ± 0.27 | 0/3 |
| 8 | +PMC | 0.67 | 0.813 ± 0.009 | 0.242 ± 0.035 | 0.805 ± 0.002 | 1.60 ± 0.24 | 0/3 |
| 8 | +PMC+BGA | 0.67 | 0.813 ± 0.005 | 0.259 ± 0.034 | 0.806 ± 0.004 | 1.48 ± 0.11 | 0/3 |
| 8 | Full (+IPP) | 0.67 | 0.810 ± 0.006 | 0.262 ± 0.017 | 0.802 ± 0.011 | 1.32 ± 0.09 | 0/3 |
| 16 | Dice only | 0.73 | 0.813 ± 0.002 | 0.226 ± 0.008 | 0.802 ± 0.006 | 1.54 ± 0.12 | 0/3 |
| 16 | +PMC | 0.73 | 0.813 ± 0.007 | 0.244 ± 0.035 | 0.811 ± 0.006 | 1.58 ± 0.24 | 0/3 |
| 16 | +PMC+BGA | 0.73 | 0.817 ± 0.003 | 0.237 ± 0.015 | 0.807 ± 0.008 | 1.55 ± 0.12 | 0/3 |
| 16 | Full (+IPP) | 0.73 | 0.813 ± 0.002 | 0.279 ± 0.054 | 0.801 ± 0.013 | 1.29 ± 0.18 | 0/3 |
| 32 | Dice only | 0.85 | 0.674 ± 0.126 | 0.230 ± 0.037 | 0.770 ± 0.043 | 9.23 ± 6.63 | 2/3 (seed 42, 43) |
| 32 | +PMC | 0.85 | 0.817 ± 0.004 | 0.265 ± 0.017 | 0.814 ± 0.007 | 1.58 ± 0.20 | 0/3 |
| 32 | +PMC+BGA | 0.85 | 0.818 ± 0.007 | 0.266 ± 0.019 | 0.814 ± 0.003 | 1.46 ± 0.16 | 0/3 |
| 32 | Full (+IPP) | 0.85 | 0.687 ± 0.234 | 0.283 ± 0.020 | 0.794 ± 0.023 | 1.76 ± 0.53 | 1/3 (seed 43) |

**Table 10:** MobileNet-UNet: all losses at all ranks (mean ± SD over seeds).

| Rank | Loss | Params (M) | Dice | AJI | BF1 | IPE (%) | Runs that lost a class |
|---|---|---|---|---|---|---|---|
| 2 | Dice only | 5.47 | 0.416 ± 0.004 | 0.579 ± 0.027 | 0.790 ± 0.004 | 17.98 ± 0.40 | 3/3 (seed 42, 43, 44) |
| 2 | +PMC | 5.47 | 0.413 ± 0.181 | 0.570 ± 0.050 | 0.781 ± 0.030 | 19.88 ± 6.25 | 3/3 (seed 42, 43, 44) |
| 2 | +PMC+BGA | 5.47 | 0.549 ± 0.112 | 0.647 ± 0.027 | 0.803 ± 0.015 | 12.40 ± 6.34 | 3/3 (seed 42, 43, 44) |
| 2 | Full (+IPP) | 5.47 | 0.410 ± 0.004 | 0.534 ± 0.029 | 0.816 ± 0.006 | 2.61 ± 0.23 | 3/3 (seed 42, 43, 44) |
| 4 | Dice only | 5.51 | 0.482 ± 0.109 | 0.629 ± 0.017 | 0.799 ± 0.014 | 16.52 ± 2.83 | 3/3 (seed 42, 43, 44) |
| 4 | +PMC | 5.51 | 0.417 ± 0.185 | 0.595 ± 0.075 | 0.781 ± 0.032 | 19.39 ± 6.20 | 3/3 (seed 42, 43, 44) |
| 4 | +PMC+BGA | 5.51 | 0.490 ± 0.224 | 0.560 ± 0.060 | 0.790 ± 0.043 | 14.70 ± 10.39 | 3/3 (seed 42, 43, 44) |
| 4 | Full (+IPP) | 5.51 | 0.406 ± 0.002 | 0.560 ± 0.035 | 0.805 ± 0.006 | 2.50 ± 0.19 | 3/3 (seed 42, 43, 44) |
| 8 | Dice only | 5.58 | 0.485 ± 0.294 | 0.623 ± 0.111 | 0.814 ± 0.054 | 15.48 ± 12.73 | 2/3 (seed 42, 44) |
| 8 | +PMC | 5.58 | 0.414 ± 0.181 | 0.585 ± 0.071 | 0.782 ± 0.022 | 19.79 ± 6.71 | 3/3 (seed 42, 43, 44) |
| 8 | +PMC+BGA | 5.58 | 0.544 ± 0.292 | 0.602 ± 0.125 | 0.812 ± 0.052 | 13.96 ± 12.46 | 2/3 (seed 42, 44) |
| 8 | Full (+IPP) | 5.58 | 0.412 ± 0.004 | 0.551 ± 0.053 | 0.814 ± 0.003 | 2.60 ± 0.07 | 3/3 (seed 42, 43, 44) |
| 16 | Dice only | 5.73 | 0.550 ± 0.297 | 0.619 ± 0.089 | 0.815 ± 0.054 | 14.18 ± 12.75 | 2/3 (seed 42, 44) |
| 16 | +PMC | 5.73 | 0.413 ± 0.176 | 0.558 ± 0.098 | 0.781 ± 0.018 | 20.13 ± 5.77 | 3/3 (seed 42, 43, 44) |
| 16 | +PMC+BGA | 5.73 | 0.612 ± 0.197 | 0.642 ± 0.033 | 0.816 ± 0.047 | 11.50 ± 8.96 | 2/3 (seed 42, 43) |
| 16 | Full (+IPP) | 5.73 | 0.414 ± 0.002 | 0.505 ± 0.012 | 0.805 ± 0.009 | 3.09 ± 0.65 | 3/3 (seed 42, 43, 44) |
| 32 | Dice only | 6.04 | 0.359 ± 0.110 | 0.592 ± 0.023 | 0.783 ± 0.020 | 20.83 ± 5.37 | 3/3 (seed 42, 43, 44) |
| 32 | +PMC | 6.04 | 0.486 ± 0.102 | 0.599 ± 0.033 | 0.800 ± 0.013 | 15.96 ± 2.29 | 3/3 (seed 42, 43, 44) |
| 32 | +PMC+BGA | 6.04 | 0.418 ± 0.185 | 0.568 ± 0.078 | 0.788 ± 0.027 | 19.54 ± 6.58 | 3/3 (seed 42, 43, 44) |
| 32 | Full (+IPP) | 6.04 | 0.476 ± 0.108 | 0.541 ± 0.049 | 0.824 ± 0.014 | 5.83 ± 5.75 | 3/3 (seed 42, 43, 44) |



# C Per-class Dice and per-class IPE

**Table 11:** EdgeSAM: per-class Dice and per-class IPE (mean ± SD over seeds). A class IPE far above 100% means over-estimation of that class.

| Rank | Loss | Dice Disco | Dice Sphero | Dice Echi | Dice Stoma | IPE Disco (%) | IPE Sphero (%) | IPE Echi (%) | IPE Stoma (%) |
|---|---|---|---|---|---|---|---|---|---|
| 2 | Dice only | 0.891 ± 0.003 | 0.186 ± 0.036 | 0.655 ± 0.011 | 0.573 ± 0.009 | 7.16 ± 0.20 | 77.56 ± 14.80 | 29.56 ± 2.94 | 45.34 ± 6.10 |
| 2 | +PMC | 0.886 ± 0.006 | 0.767 ± 0.013 | 0.633 ± 0.009 | 0.557 ± 0.019 | 7.83 ± 0.81 | 24.42 ± 1.97 | 35.59 ± 6.18 | 48.24 ± 1.91 |
| 2 | +PMC+BGA | 0.888 ± 0.003 | 0.777 ± 0.015 | 0.635 ± 0.004 | 0.569 ± 0.020 | 8.03 ± 0.24 | 25.94 ± 2.71 | 32.98 ± 1.12 | 50.31 ± 5.56 |
| 2 | Full (+IPP) | 0.882 ± 0.008 | 0.497 ± 0.431 | 0.616 ± 0.010 | 0.506 ± 0.013 | 13.37 ± 10.72 | 51.20 ± 42.28 | 36.78 ± 2.74 | 60.32 ± 6.51 |
| 4 | Dice only | 0.890 ± 0.003 | 0.331 ± 0.360 | 0.649 ± 0.016 | 0.571 ± 0.030 | 7.17 ± 0.17 | 66.14 ± 33.21 | 31.07 ± 3.03 | 51.70 ± 2.56 |
| 4 | +PMC | 0.892 ± 0.003 | 0.772 ± 0.019 | 0.655 ± 0.013 | 0.588 ± 0.006 | 7.41 ± 0.11 | 25.70 ± 4.44 | 30.41 ± 0.78 | 47.43 ± 3.54 |
| 4 | +PMC+BGA | 0.890 ± 0.005 | 0.770 ± 0.008 | 0.641 ± 0.014 | 0.593 ± 0.024 | 7.51 ± 0.22 | 21.76 ± 3.98 | 32.79 ± 2.30 | 45.42 ± 6.85 |
| 4 | Full (+IPP) | 0.891 ± 0.002 | 0.766 ± 0.013 | 0.612 ± 0.007 | 0.544 ± 0.025 | 6.46 ± 0.29 | 30.37 ± 0.67 | 35.55 ± 1.80 | 50.42 ± 6.04 |
| 8 | Dice only | 0.893 ± 0.003 | 0.343 ± 0.373 | 0.668 ± 0.013 | 0.592 ± 0.006 | 6.54 ± 0.08 | 64.24 ± 30.77 | 28.72 ± 3.36 | 49.40 ± 1.01 |
| 8 | +PMC | 0.892 ± 0.005 | 0.524 ± 0.454 | 0.669 ± 0.011 | 0.601 ± 0.016 | 7.39 ± 0.32 | 49.49 ± 43.74 | 29.71 ± 0.80 | 47.10 ± 1.86 |
| 8 | +PMC+BGA | 0.892 ± 0.004 | 0.783 ± 0.013 | 0.654 ± 0.015 | 0.612 ± 0.020 | 7.65 ± 0.71 | 22.12 ± 1.20 | 31.53 ± 2.03 | 46.65 ± 5.82 |
| 8 | Full (+IPP) | 0.882 ± 0.012 | 0.513 ± 0.444 | 0.628 ± 0.006 | 0.563 ± 0.019 | 22.49 ± 27.34 | 51.52 ± 42.04 | 34.26 ± 2.62 | 49.58 ± 7.29 |
| 16 | Dice only | 0.898 ± 0.002 | 0.273 ± 0.434 | 0.676 ± 0.009 | 0.636 ± 0.028 | 6.00 ± 0.35 | 66.86 ± 41.60 | 31.46 ± 1.81 | 48.84 ± 4.42 |
| 16 | +PMC | 0.894 ± 0.004 | 0.783 ± 0.001 | 0.681 ± 0.010 | 0.604 ± 0.029 | 6.84 ± 0.89 | 23.04 ± 2.49 | 29.29 ± 4.93 | 47.69 ± 1.37 |
| 16 | +PMC+BGA | 0.896 ± 0.003 | 0.778 ± 0.011 | 0.676 ± 0.009 | 0.617 ± 0.005 | 6.25 ± 0.37 | 25.79 ± 4.44 | 27.02 ± 1.49 | 49.06 ± 3.98 |
| 16 | Full (+IPP) | 0.890 ± 0.007 | 0.521 ± 0.451 | 0.659 ± 0.015 | 0.591 ± 0.027 | 14.14 ± 13.42 | 49.32 ± 43.89 | 32.19 ± 3.53 | 55.86 ± 7.64 |
| 32 | Dice only | 0.898 ± 0.002 | 0.272 ± 0.439 | 0.675 ± 0.004 | 0.630 ± 0.013 | 6.96 ± 0.35 | 71.17 ± 38.14 | 30.83 ± 3.34 | 43.59 ± 2.24 |
| 32 | +PMC | 0.896 ± 0.003 | 0.522 ± 0.452 | 0.678 ± 0.003 | 0.625 ± 0.007 | 6.50 ± 0.17 | 48.36 ± 44.80 | 27.96 ± 1.80 | 47.80 ± 4.27 |
| 32 | +PMC+BGA | 0.896 ± 0.003 | 0.781 ± 0.005 | 0.678 ± 0.005 | 0.627 ± 0.024 | 6.35 ± 1.05 | 24.85 ± 4.47 | 30.42 ± 0.75 | 44.64 ± 3.27 |
| 32 | Full (+IPP) | 0.895 ± 0.001 | 0.789 ± 0.005 | 0.655 ± 0.010 | 0.614 ± 0.019 | 6.44 ± 0.32 | 24.24 ± 4.23 | 28.74 ± 0.65 | 52.21 ± 2.71 |

**Table 12:** MobileSAM: per-class Dice and per-class IPE (mean ± SD over seeds). A class IPE far above 100% means over-estimation of that class.

| Rank | Loss | Dice Disco | Dice Sphero | Dice Echi | Dice Stoma | IPE Disco (%) | IPE Sphero (%) | IPE Echi (%) | IPE Stoma (%) |
|---|---|---|---|---|---|---|---|---|---|
| 2 | Dice only | 0.923 ± 0.000 | 0.565 ± 0.489 | 0.766 ± 0.011 | 0.710 ± 0.008 | 6.87 ± 0.66 | 49.67 ± 43.60 | 24.81 ± 5.46 | 47.68 ± 2.15 |
| 2 | +PMC | 0.923 ± 0.002 | 0.846 ± 0.006 | 0.753 ± 0.015 | 0.697 ± 0.027 | 7.15 ± 0.16 | 25.06 ± 3.47 | 25.07 ± 1.71 | 48.89 ± 5.70 |
| 2 | +PMC+BGA | 0.922 ± 0.001 | 0.572 ± 0.495 | 0.770 ± 0.010 | 0.698 ± 0.006 | 6.63 ± 0.57 | 48.80 ± 44.38 | 23.06 ± 1.25 | 49.01 ± 5.16 |
| 2 | Full (+IPP) | 0.924 ± 0.000 | 0.848 ± 0.002 | 0.767 ± 0.001 | 0.694 ± 0.013 | 6.91 ± 0.41 | 23.81 ± 3.39 | 24.33 ± 0.33 | 50.22 ± 6.81 |
| 4 | Dice only | 0.924 ± 0.002 | 0.849 ± 0.006 | 0.761 ± 0.008 | 0.713 ± 0.008 | 6.89 ± 0.78 | 22.70 ± 2.82 | 26.74 ± 3.21 | 50.45 ± 1.69 |
| 4 | +PMC | 0.923 ± 0.001 | 0.850 ± 0.009 | 0.755 ± 0.017 | 0.695 ± 0.018 | 7.24 ± 0.43 | 23.25 ± 5.68 | 26.08 ± 2.03 | 49.27 ± 3.15 |
| 4 | +PMC+BGA | 0.924 ± 0.001 | 0.564 ± 0.488 | 0.763 ± 0.003 | 0.711 ± 0.004 | 6.62 ± 0.52 | 49.16 ± 44.06 | 29.11 ± 4.38 | 47.47 ± 5.91 |
| 4 | Full (+IPP) | 0.923 ± 0.002 | 0.855 ± 0.005 | 0.773 ± 0.003 | 0.691 ± 0.021 | 7.13 ± 0.29 | 19.38 ± 0.95 | 24.45 ± 0.22 | 51.66 ± 3.83 |
| 8 | Dice only | 0.924 ± 0.003 | 0.849 ± 0.005 | 0.758 ± 0.010 | 0.713 ± 0.007 | 6.79 ± 1.07 | 20.82 ± 4.09 | 27.13 ± 1.29 | 48.45 ± 4.83 |
| 8 | +PMC | 0.924 ± 0.002 | 0.852 ± 0.007 | 0.775 ± 0.009 | 0.701 ± 0.027 | 6.51 ± 0.44 | 23.74 ± 1.83 | 25.36 ± 0.82 | 50.49 ± 3.23 |
| 8 | +PMC+BGA | 0.924 ± 0.001 | 0.854 ± 0.004 | 0.766 ± 0.003 | 0.710 ± 0.018 | 7.01 ± 0.64 | 20.31 ± 3.03 | 26.44 ± 0.98 | 53.94 ± 3.93 |
| 8 | Full (+IPP) | 0.923 ± 0.003 | 0.851 ± 0.004 | 0.760 ± 0.006 | 0.708 ± 0.020 | 7.15 ± 0.62 | 22.12 ± 1.15 | 28.75 ± 4.22 | 55.15 ± 3.37 |
| 16 | Dice only | 0.925 ± 0.001 | 0.852 ± 0.001 | 0.766 ± 0.018 | 0.711 ± 0.009 | 6.37 ± 0.52 | 21.56 ± 0.96 | 29.61 ± 1.40 | 46.47 ± 1.05 |
| 16 | +PMC | 0.926 ± 0.002 | 0.848 ± 0.012 | 0.765 ± 0.007 | 0.714 ± 0.010 | 6.66 ± 0.80 | 21.66 ± 4.66 | 24.57 ± 1.09 | 50.66 ± 4.58 |
| 16 | +PMC+BGA | 0.926 ± 0.001 | 0.853 ± 0.004 | 0.770 ± 0.005 | 0.719 ± 0.013 | 6.35 ± 0.75 | 24.10 ± 3.47 | 27.58 ± 1.89 | 49.84 ± 3.78 |
| 16 | Full (+IPP) | 0.923 ± 0.003 | 0.855 ± 0.002 | 0.777 ± 0.007 | 0.696 ± 0.009 | 6.51 ± 0.22 | 21.93 ± 1.46 | 26.62 ± 2.71 | 52.74 ± 3.08 |
| 32 | Dice only | 0.924 ± 0.002 | 0.284 ± 0.492 | 0.763 ± 0.005 | 0.722 ± 0.008 | 6.35 ± 0.35 | 74.53 ± 44.12 | 26.84 ± 2.95 | 50.60 ± 2.77 |
| 32 | +PMC | 0.926 ± 0.001 | 0.859 ± 0.006 | 0.770 ± 0.003 | 0.713 ± 0.024 | 6.47 ± 1.02 | 19.79 ± 2.96 | 24.64 ± 0.82 | 52.39 ± 3.91 |
| 32 | +PMC+BGA | 0.925 ± 0.000 | 0.858 ± 0.006 | 0.773 ± 0.012 | 0.714 ± 0.011 | 7.03 ± 0.57 | 20.05 ± 1.05 | 26.42 ± 0.79 | 46.61 ± 3.51 |
| 32 | Full (+IPP) | 0.913 ± 0.020 | 0.574 ± 0.497 | 0.777 ± 0.004 | 0.485 ± 0.420 | 27.69 ± 37.28 | 46.57 ± 46.30 | 27.97 ± 4.24 | 65.55 ± 29.84 |

**Table 13:** MobileNet-UNet: per-class Dice and per-class IPE (mean ± SD over seeds). A class IPE far above 100% means over-estimation of that class.

| Rank | Loss | Dice Disco | Dice Sphero | Dice Echi | Dice Stoma | IPE Disco (%) | IPE Sphero (%) | IPE Echi (%) | IPE Stoma (%) |
|---|---|---|---|---|---|---|---|---|---|
| 2 | Dice only | 0.926 ± 0.001 | 0.000 ± 0.000 | 0.739 ± 0.013 | 0.000 ± 0.000 | 6.98 ± 0.23 | 100.00 ± 0.00 | 30.97 ± 3.02 | 100.00 ± 0.00 |
| 2 | +PMC | 0.926 ± 0.001 | 0.000 ± 0.000 | 0.495 ± 0.428 | 0.233 ± 0.403 | 6.59 ± 0.97 | 100.00 ± 0.00 | 54.86 ± 39.10 | 84.39 ± 27.04 |
| 2 | +PMC+BGA | 0.926 ± 0.001 | 0.287 ± 0.497 | 0.752 ± 0.007 | 0.232 ± 0.402 | 6.44 ± 0.17 | 72.56 ± 47.53 | 30.92 ± 4.41 | 83.27 ± 28.97 |
| 2 | Full (+IPP) | 0.889 ± 0.001 | 0.000 ± 0.000 | 0.753 ± 0.016 | 0.000 ± 0.000 | 81.96 ± 2.78 | 100.00 ± 0.00 | 30.58 ± 3.20 | 100.00 ± 0.00 |
| 4 | Dice only | 0.927 ± 0.002 | 0.000 ± 0.000 | 0.760 ± 0.019 | 0.240 ± 0.417 | 6.61 ± 0.35 | 100.00 ± 0.00 | 27.92 ± 2.58 | 83.57 ± 28.45 |
| 4 | +PMC | 0.927 ± 0.001 | 0.000 ± 0.000 | 0.502 ± 0.435 | 0.240 ± 0.416 | 6.47 ± 1.90 | 100.00 ± 0.00 | 54.04 ± 40.07 | 83.46 ± 28.65 |
| 4 | +PMC+BGA | 0.926 ± 0.003 | 0.286 ± 0.495 | 0.505 ± 0.438 | 0.242 ± 0.419 | 6.79 ± 1.32 | 73.14 ± 46.52 | 53.06 ± 40.66 | 82.47 ± 30.37 |
| 4 | Full (+IPP) | 0.889 ± 0.001 | 0.000 ± 0.000 | 0.737 ± 0.005 | 0.000 ± 0.000 | 82.49 ± 2.05 | 100.00 ± 0.00 | 33.13 ± 1.25 | 100.00 ± 0.00 |
| 8 | Dice only | 0.928 ± 0.002 | 0.282 ± 0.489 | 0.496 ± 0.430 | 0.233 ± 0.404 | 6.51 ± 1.17 | 72.90 ± 46.94 | 52.53 ± 41.11 | 83.66 ± 28.30 |
| 8 | +PMC | 0.926 ± 0.002 | 0.000 ± 0.000 | 0.500 ± 0.433 | 0.229 ± 0.397 | 6.89 ± 0.55 | 100.00 ± 0.00 | 54.30 ± 39.58 | 82.44 ± 30.41 |
| 8 | +PMC+BGA | 0.926 ± 0.001 | 0.284 ± 0.491 | 0.497 ± 0.431 | 0.470 ± 0.407 | 6.37 ± 0.70 | 73.21 ± 46.40 | 56.51 ± 37.83 | 67.92 ± 27.99 |
| 8 | Full (+IPP) | 0.891 ± 0.002 | 0.000 ± 0.000 | 0.756 ± 0.015 | 0.000 ± 0.000 | 79.49 ± 4.87 | 100.00 ± 0.00 | 28.16 ± 3.43 | 100.00 ± 0.00 |
| 16 | Dice only | 0.930 ± 0.001 | 0.283 ± 0.491 | 0.498 ± 0.431 | 0.491 ± 0.425 | 6.13 ± 1.07 | 74.25 ± 44.60 | 51.69 ± 41.84 | 63.68 ± 31.46 |
| 16 | +PMC | 0.927 ± 0.002 | 0.000 ± 0.000 | 0.494 ± 0.428 | 0.230 ± 0.399 | 6.41 ± 0.66 | 100.00 ± 0.00 | 54.41 ± 39.49 | 80.28 ± 34.15 |
| 16 | +PMC+BGA | 0.927 ± 0.002 | 0.284 ± 0.492 | 0.754 ± 0.007 | 0.481 ± 0.417 | 6.68 ± 0.28 | 73.51 ± 45.88 | 27.10 ± 3.76 | 64.49 ± 30.94 |
| 16 | Full (+IPP) | 0.892 ± 0.001 | 0.000 ± 0.000 | 0.764 ± 0.006 | 0.000 ± 0.000 | 76.96 ± 4.24 | 100.00 ± 0.00 | 24.82 ± 2.80 | 100.00 ± 0.00 |
| 32 | Dice only | 0.928 ± 0.002 | 0.000 ± 0.000 | 0.508 ± 0.440 | 0.000 ± 0.000 | 6.19 ± 0.96 | 100.00 ± 0.00 | 51.76 ± 41.80 | 100.00 ± 0.00 |
| 32 | +PMC | 0.927 ± 0.001 | 0.000 ± 0.000 | 0.772 ± 0.015 | 0.243 ± 0.421 | 6.10 ± 1.08 | 100.00 ± 0.00 | 27.30 ± 5.15 | 83.24 ± 29.03 |
| 32 | +PMC+BGA | 0.928 ± 0.001 | 0.000 ± 0.000 | 0.501 ± 0.434 | 0.244 ± 0.422 | 6.67 ± 0.55 | 100.00 ± 0.00 | 51.18 ± 42.28 | 80.99 ± 32.93 |
| 32 | Full (+IPP) | 0.903 ± 0.022 | 0.000 ± 0.000 | 0.767 ± 0.003 | 0.236 ± 0.408 | 54.55 ± 42.15 | 100.00 ± 0.00 | 29.33 ± 2.54 | 86.08 ± 24.10 |

# D Paired differences (same seed in both runs)

**Table 14:** EdgeSAM: paired differences. IPE in percentage points, Dice and AJI as fractions.

| Rank | Comparison (b − a) | Metric | Mean difference | 95% $t$-interval | b higher in |
|---|---|---|---|---|---|
| 2 | +PMC − Dice only | IPE (pp) | −9.15 | [−11.50, −6.81] | 0 of 3 seeds |
| 2 | +PMC − Dice only | Dice | 0.134 | [0.115, 0.153] | 3 of 3 seeds |
| 2 | +PMC − Dice only | AJI | 0.063 | [−0.021, 0.147] | 3 of 3 seeds |
| 2 | +PMC+BGA − Dice only | IPE (pp) | −9.54 | [−14.30, −4.78] | 0 of 3 seeds |
| 2 | +PMC+BGA − Dice only | Dice | 0.141 | [0.094, 0.188] | 3 of 3 seeds |
| 2 | +PMC+BGA − Dice only | AJI | 0.074 | [0.021, 0.128] | 3 of 3 seeds |
| 2 | Full (+IPP) − Dice only | IPE (pp) | −9.97 | [−18.58, −1.35] | 0 of 3 seeds |
| 2 | Full (+IPP) − Dice only | Dice | 0.049 | [−0.245, 0.344] | 2 of 3 seeds |
| 2 | Full (+IPP) − Dice only | AJI | −0.062 | [−0.157, 0.033] | 0 of 3 seeds |
| 2 | Full (+IPP) − +PMC+BGA | IPE (pp) | −0.43 | [−4.29, 3.43] | 1 of 3 seeds |
| 2 | Full (+IPP) − +PMC+BGA | Dice | −0.092 | [−0.346, 0.163] | 0 of 3 seeds |
| 2 | Full (+IPP) − +PMC+BGA | AJI | −0.136 | [−0.283, 0.010] | 0 of 3 seeds |
| 4 | +PMC − Dice only | IPE (pp) | −6.28 | [−16.49, 3.93] | 0 of 3 seeds |
| 4 | +PMC − Dice only | Dice | 0.117 | [−0.090, 0.324] | 3 of 3 seeds |
| 4 | +PMC − Dice only | AJI | 0.042 | [−0.173, 0.257] | 2 of 3 seeds |
| 4 | +PMC+BGA − Dice only | IPE (pp) | −5.98 | [−15.60, 3.64] | 0 of 3 seeds |
| 4 | +PMC+BGA − Dice only | Dice | 0.113 | [−0.093, 0.320] | 3 of 3 seeds |

| Rank | Comparison (b − a) | Metric | Mean difference | 95% $t$-interval | b higher in |
|---|---|---|---|---|---|
| 4 | +PMC+BGA − Dice only | AJI | 0.071 | [−0.093, 0.234] | 3 of 3 seeds |
| 4 | Full (+IPP) − Dice only | IPE (pp) | −7.48 | [−19.64, 4.67] | 0 of 3 seeds |
| 4 | Full (+IPP) − Dice only | Dice | 0.093 | [−0.107, 0.293] | 3 of 3 seeds |
| 4 | Full (+IPP) − Dice only | AJI | −0.069 | [−0.193, 0.055] | 0 of 3 seeds |
| 4 | Full (+IPP) − +PMC+BGA | IPE (pp) | −1.50 | [−4.04, 1.03] | 0 of 3 seeds |
| 4 | Full (+IPP) − +PMC+BGA | Dice | −0.020 | [−0.028, −0.012] | 0 of 3 seeds |
| 4 | Full (+IPP) − +PMC+BGA | AJI | −0.140 | [−0.193, −0.086] | 0 of 3 seeds |
| 8 | +PMC − Dice only | IPE (pp) | −2.84 | [−16.76, 11.07] | 1 of 3 seeds |
| 8 | +PMC − Dice only | Dice | 0.048 | [−0.253, 0.349] | 2 of 3 seeds |
| 8 | +PMC − Dice only | AJI | 0.052 | [−0.139, 0.243] | 3 of 3 seeds |
| 8 | +PMC+BGA − Dice only | IPE (pp) | −6.57 | [−17.30, 4.16] | 0 of 3 seeds |
| 8 | +PMC+BGA − Dice only | Dice | 0.111 | [−0.102, 0.325] | 3 of 3 seeds |
| 8 | +PMC+BGA − Dice only | AJI | 0.057 | [−0.107, 0.220] | 2 of 3 seeds |
| 8 | Full (+IPP) − Dice only | IPE (pp) | −7.01 | [−19.68, 5.66] | 0 of 3 seeds |
| 8 | Full (+IPP) − Dice only | Dice | 0.023 | [−0.306, 0.352] | 1 of 3 seeds |
| 8 | Full (+IPP) − Dice only | AJI | −0.084 | [−0.176, 0.007] | 0 of 3 seeds |
| 8 | Full (+IPP) − +PMC+BGA | IPE (pp) | −0.44 | [−4.77, 3.90] | 2 of 3 seeds |
| 8 | Full (+IPP) − +PMC+BGA | Dice | −0.089 | [−0.357, 0.179] | 0 of 3 seeds |
| 8 | Full (+IPP) − +PMC+BGA | AJI | −0.141 | [−0.216, −0.066] | 0 of 3 seeds |
| 16 | +PMC − Dice only | IPE (pp) | −7.34 | [−21.59, 6.91] | 0 of 3 seeds |
| 16 | +PMC − Dice only | Dice | 0.120 | [−0.114, 0.354] | 3 of 3 seeds |
| 16 | +PMC − Dice only | AJI | 0.083 | [0.004, 0.163] | 3 of 3 seeds |
| 16 | +PMC+BGA − Dice only | IPE (pp) | −7.62 | [−20.88, 5.64] | 0 of 3 seeds |
| 16 | +PMC+BGA − Dice only | Dice | 0.121 | [−0.121, 0.364] | 3 of 3 seeds |
| 16 | +PMC+BGA − Dice only | AJI | 0.025 | [−0.099, 0.148] | 2 of 3 seeds |
| 16 | Full (+IPP) − Dice only | IPE (pp) | −6.39 | [−19.07, 6.29] | 0 of 3 seeds |
| 16 | Full (+IPP) − Dice only | Dice | 0.045 | [−0.229, 0.319] | 1 of 3 seeds |
| 16 | Full (+IPP) − Dice only | AJI | −0.037 | [−0.098, 0.023] | 0 of 3 seeds |
| 16 | Full (+IPP) − +PMC+BGA | IPE (pp) | 1.23 | [−5.74, 8.21] | 2 of 3 seeds |
| 16 | Full (+IPP) − +PMC+BGA | Dice | −0.077 | [−0.353, 0.199] | 1 of 3 seeds |
| 16 | Full (+IPP) − +PMC+BGA | AJI | −0.062 | [−0.139, 0.015] | 0 of 3 seeds |
| 32 | +PMC − Dice only | IPE (pp) | −3.75 | [−20.14, 12.65] | 1 of 3 seeds |
| 32 | +PMC − Dice only | Dice | 0.061 | [−0.192, 0.314] | 3 of 3 seeds |
| 32 | +PMC − Dice only | AJI | 0.060 | [−0.072, 0.192] | 3 of 3 seeds |
| 32 | +PMC+BGA − Dice only | IPE (pp) | −7.56 | [−23.01, 7.89] | 0 of 3 seeds |
| 32 | +PMC+BGA − Dice only | Dice | 0.127 | [−0.128, 0.382] | 3 of 3 seeds |
| 32 | +PMC+BGA − Dice only | AJI | 0.077 | [−0.001, 0.156] | 3 of 3 seeds |
| 32 | Full (+IPP) − Dice only | IPE (pp) | −8.18 | [−23.52, 7.15] | 0 of 3 seeds |
| 32 | Full (+IPP) − Dice only | Dice | 0.119 | [−0.166, 0.404] | 2 of 3 seeds |
| 32 | Full (+IPP) − Dice only | AJI | 0.007 | [−0.011, 0.024] | 3 of 3 seeds |
| 32 | Full (+IPP) − +PMC+BGA | IPE (pp) | −0.63 | [−0.75, −0.51] | 0 of 3 seeds |
| 32 | Full (+IPP) − +PMC+BGA | Dice | −0.008 | [−0.039, 0.023] | 1 of 3 seeds |
| 32 | Full (+IPP) − +PMC+BGA | AJI | −0.070 | [−0.137, −0.004] | 0 of 3 seeds |
| – | Full (+IPP), r = 4 − Full (+IPP), r = 2 | Dice | 0.078 | [−0.169, 0.324] | 3 of 3 seeds |
| – | Full (+IPP), r = 4 − Full (+IPP), r = 2 | AJI | 0.003 | [−0.056, 0.063] | 2 of 3 seeds |
| – | Full (+IPP), r = 4 − Full (+IPP), r = 2 | IPE (pp) | −0.98 | [−5.44, 3.49] | 1 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 2 | Dice | 0.021 | [0.005, 0.037] | 3 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 2 | AJI | 0.001 | [−0.089, 0.090] | 1 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 2 | IPE (pp) | −0.64 | [−3.13, 1.84] | 1 of 3 seeds |

| Rank | Comparison (b − a) | Metric | Mean difference | 95% $t$-interval | b higher in |
|---|---|---|---|---|---|
| – | Full (+IPP), r = 16 − Full (+IPP), r = 2 | Dice | 0.040 | [0.019, 0.060] | 3 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 2 | AJI | 0.028 | [−0.002, 0.059] | 3 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 2 | IPE (pp) | 0.21 | [−1.95, 2.37] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 2 | Dice | 0.113 | [−0.161, 0.386] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 2 | AJI | 0.087 | [0.054, 0.119] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 2 | IPE (pp) | −1.44 | [−6.10, 3.23] | 0 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 4 | Dice | −0.057 | [−0.318, 0.204] | 1 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 4 | AJI | −0.003 | [−0.063, 0.057] | 1 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 4 | IPE (pp) | 0.34 | [−1.97, 2.64] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 4 | Dice | −0.038 | [−0.301, 0.224] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 4 | AJI | 0.025 | [−0.016, 0.066] | 3 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 4 | IPE (pp) | 1.19 | [−5.33, 7.70] | 1 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 4 | Dice | 0.035 | [0.002, 0.067] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 4 | AJI | 0.083 | [0.051, 0.115] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 4 | IPE (pp) | −0.46 | [−0.74, −0.18] | 0 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 8 | Dice | 0.019 | [0.013, 0.024] | 3 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 8 | AJI | 0.028 | [−0.032, 0.087] | 3 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 8 | IPE (pp) | 0.85 | [−3.47, 5.18] | 1 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 8 | Dice | 0.091 | [−0.196, 0.379] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 8 | AJI | 0.086 | [0.026, 0.146] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 8 | IPE (pp) | −0.80 | [−3.22, 1.62] | 0 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 16 | Dice | 0.073 | [−0.215, 0.361] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 16 | AJI | 0.058 | [0.048, 0.068] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 16 | IPE (pp) | −1.65 | [−8.33, 5.04] | 1 of 3 seeds |



**Table 15:** MobileSAM: paired differences. IPE in percentage points, Dice and AJI as fractions.

| Rank | Comparison (b − a) | Metric | Mean difference | 95% $t$-interval | b higher in |
|---|---|---|---|---|---|
| 2 | +PMC − Dice only | IPE (pp) | −3.80 | [−20.03, 12.43] | 0 of 3 seeds |
| 2 | +PMC − Dice only | Dice | 0.064 | [−0.234, 0.362] | 2 of 3 seeds |
| 2 | +PMC − Dice only | AJI | 0.010 | [−0.097, 0.116] | 2 of 3 seeds |
| 2 | +PMC+BGA − Dice only | IPE (pp) | −0.04 | [−0.17, 0.09] | 1 of 3 seeds |
| 2 | +PMC+BGA − Dice only | Dice | −0.000 | [−0.005, 0.005] | 2 of 3 seeds |
| 2 | +PMC+BGA − Dice only | AJI | 0.019 | [−0.023, 0.060] | 3 of 3 seeds |
| 2 | Full (+IPP) − Dice only | IPE (pp) | −4.06 | [−20.36, 12.24] | 0 of 3 seeds |
| 2 | Full (+IPP) − Dice only | Dice | 0.067 | [−0.233, 0.368] | 1 of 3 seeds |
| 2 | Full (+IPP) − Dice only | AJI | 0.045 | [−0.015, 0.106] | 3 of 3 seeds |
| 2 | Full (+IPP) − +PMC+BGA | IPE (pp) | −4.02 | [−20.23, 12.19] | 0 of 3 seeds |
| 2 | Full (+IPP) − +PMC+BGA | Dice | 0.068 | [−0.238, 0.373] | 1 of 3 seeds |
| 2 | Full (+IPP) − +PMC+BGA | AJI | 0.027 | [−0.017, 0.071] | 3 of 3 seeds |
| 4 | +PMC − Dice only | IPE (pp) | 0.06 | [−0.06, 0.18] | 3 of 3 seeds |
| 4 | +PMC − Dice only | Dice | −0.006 | [−0.024, 0.013] | 1 of 3 seeds |
| 4 | +PMC − Dice only | AJI | −0.032 | [−0.075, 0.011] | 0 of 3 seeds |
| 4 | +PMC+BGA − Dice only | IPE (pp) | 3.79 | [−12.24, 19.83] | 2 of 3 seeds |
| 4 | +PMC+BGA − Dice only | Dice | −0.071 | [−0.377, 0.235] | 1 of 3 seeds |
| 4 | +PMC+BGA − Dice only | AJI | −0.021 | [−0.050, 0.009] | 0 of 3 seeds |

| Rank | Comparison (b − a) | Metric | Mean difference | 95% $t$-interval | b higher in |
|---|---|---|---|---|---|
| 4 | Full (+IPP) − Dice only | IPE (pp) | −0.10 | [−0.30, 0.11] | 0 of 3 seeds |
| 4 | Full (+IPP) − Dice only | Dice | −0.001 | [−0.016, 0.014] | 1 of 3 seeds |
| 4 | Full (+IPP) − Dice only | AJI | 0.001 | [−0.038, 0.039] | 1 of 3 seeds |
| 4 | Full (+IPP) − +PMC+BGA | IPE (pp) | −3.89 | [−20.06, 12.28] | 1 of 3 seeds |
| 4 | Full (+IPP) − +PMC+BGA | Dice | 0.070 | [−0.227, 0.367] | 2 of 3 seeds |
| 4 | Full (+IPP) − +PMC+BGA | AJI | 0.021 | [−0.003, 0.046] | 3 of 3 seeds |
| 8 | +PMC − Dice only | IPE (pp) | 0.07 | [−0.01, 0.14] | 3 of 3 seeds |
| 8 | +PMC − Dice only | Dice | 0.002 | [−0.027, 0.031] | 2 of 3 seeds |
| 8 | +PMC − Dice only | AJI | −0.006 | [−0.041, 0.029] | 1 of 3 seeds |
| 8 | +PMC+BGA − Dice only | IPE (pp) | −0.05 | [−0.46, 0.36] | 1 of 3 seeds |
| 8 | +PMC+BGA − Dice only | Dice | 0.003 | [−0.013, 0.018] | 2 of 3 seeds |
| 8 | +PMC+BGA − Dice only | AJI | 0.011 | [−0.033, 0.056] | 2 of 3 seeds |
| 8 | Full (+IPP) − Dice only | IPE (pp) | −0.21 | [−0.75, 0.33] | 1 of 3 seeds |
| 8 | Full (+IPP) − Dice only | Dice | −0.000 | [−0.017, 0.016] | 2 of 3 seeds |
| 8 | Full (+IPP) − Dice only | AJI | 0.014 | [−0.004, 0.032] | 3 of 3 seeds |
| 8 | Full (+IPP) − +PMC+BGA | IPE (pp) | −0.16 | [−0.41, 0.09] | 0 of 3 seeds |
| 8 | Full (+IPP) − +PMC+BGA | Dice | −0.003 | [−0.030, 0.024] | 1 of 3 seeds |
| 8 | Full (+IPP) − +PMC+BGA | AJI | 0.003 | [−0.057, 0.063] | 2 of 3 seeds |
| 16 | +PMC − Dice only | IPE (pp) | 0.04 | [−0.45, 0.53] | 2 of 3 seeds |
| 16 | +PMC − Dice only | Dice | 0.000 | [−0.012, 0.012] | 1 of 3 seeds |
| 16 | +PMC − Dice only | AJI | 0.018 | [−0.064, 0.099] | 2 of 3 seeds |
| 16 | +PMC+BGA − Dice only | IPE (pp) | 0.01 | [−0.11, 0.12] | 2 of 3 seeds |
| 16 | +PMC+BGA − Dice only | Dice | 0.004 | [0.001, 0.006] | 3 of 3 seeds |
| 16 | +PMC+BGA − Dice only | AJI | 0.011 | [−0.015, 0.037] | 2 of 3 seeds |
| 16 | Full (+IPP) − Dice only | IPE (pp) | −0.26 | [−0.55, 0.04] | 0 of 3 seeds |
| 16 | Full (+IPP) − Dice only | Dice | −0.000 | [−0.005, 0.004] | 1 of 3 seeds |
| 16 | Full (+IPP) − Dice only | AJI | 0.052 | [−0.062, 0.167] | 3 of 3 seeds |
| 16 | Full (+IPP) − +PMC+BGA | IPE (pp) | −0.26 | [−0.67, 0.14] | 0 of 3 seeds |
| 16 | Full (+IPP) − +PMC+BGA | Dice | −0.004 | [−0.010, 0.002] | 0 of 3 seeds |
| 16 | Full (+IPP) − +PMC+BGA | AJI | 0.042 | [−0.077, 0.161] | 2 of 3 seeds |
| 32 | +PMC − Dice only | IPE (pp) | −7.66 | [−24.51, 9.20] | 1 of 3 seeds |
| 32 | +PMC − Dice only | Dice | 0.143 | [−0.180, 0.466] | 2 of 3 seeds |
| 32 | +PMC − Dice only | AJI | 0.035 | [−0.098, 0.168] | 2 of 3 seeds |
| 32 | +PMC+BGA − Dice only | IPE (pp) | −7.77 | [−24.21, 8.67] | 0 of 3 seeds |
| 32 | +PMC+BGA − Dice only | Dice | 0.144 | [−0.175, 0.463] | 2 of 3 seeds |
| 32 | +PMC+BGA − Dice only | AJI | 0.036 | [−0.034, 0.106] | 3 of 3 seeds |
| 32 | Full (+IPP) − Dice only | IPE (pp) | −7.48 | [−23.20, 8.24] | 0 of 3 seeds |
| 32 | Full (+IPP) − Dice only | Dice | 0.014 | [−0.481, 0.509] | 2 of 3 seeds |
| 32 | Full (+IPP) − Dice only | AJI | 0.054 | [−0.083, 0.190] | 3 of 3 seeds |
| 32 | Full (+IPP) − +PMC+BGA | IPE (pp) | 0.29 | [−1.32, 1.91] | 1 of 3 seeds |
| 32 | Full (+IPP) − +PMC+BGA | Dice | −0.130 | [−0.728, 0.468] | 2 of 3 seeds |
| 32 | Full (+IPP) − +PMC+BGA | AJI | 0.017 | [−0.055, 0.090] | 2 of 3 seeds |
| – | Full (+IPP), r = 4 − Full (+IPP), r = 2 | Dice | 0.002 | [−0.012, 0.017] | 2 of 3 seeds |
| – | Full (+IPP), r = 4 − Full (+IPP), r = 2 | AJI | 0.004 | [−0.080, 0.088] | 2 of 3 seeds |
| – | Full (+IPP), r = 4 − Full (+IPP), r = 2 | IPE (pp) | 0.05 | [−0.02, 0.11] | 3 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 2 | Dice | 0.002 | [−0.007, 0.012] | 2 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 2 | AJI | −0.002 | [−0.044, 0.039] | 1 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 2 | IPE (pp) | 0.02 | [−0.19, 0.23] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 2 | Dice | 0.005 | [−0.003, 0.012] | 3 of 3 seeds |

| Rank | Comparison (b − a) | Metric | Mean difference | 95% $t$-interval | b higher in |
|---|---|---|---|---|---|
| – | Full (+IPP), r = 16 − Full (+IPP), r = 2 | AJI | 0.015 | [−0.130, 0.159] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 2 | IPE (pp) | −0.01 | [−0.25, 0.23] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 2 | Dice | −0.121 | [−0.698, 0.457] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 2 | AJI | 0.019 | [−0.013, 0.051] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 2 | IPE (pp) | 0.46 | [−1.04, 1.96] | 3 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 4 | Dice | −0.000 | [−0.024, 0.024] | 2 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 4 | AJI | −0.006 | [−0.057, 0.044] | 1 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 4 | IPE (pp) | −0.03 | [−0.18, 0.13] | 1 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 4 | Dice | 0.002 | [−0.005, 0.009] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 4 | AJI | 0.010 | [−0.071, 0.092] | 1 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 4 | IPE (pp) | −0.06 | [−0.36, 0.24] | 1 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 4 | Dice | −0.123 | [−0.714, 0.468] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 4 | AJI | 0.015 | [−0.038, 0.068] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 4 | IPE (pp) | 0.41 | [−1.03, 1.85] | 3 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 8 | Dice | 0.002 | [−0.015, 0.019] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 8 | AJI | 0.017 | [−0.086, 0.120] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 8 | IPE (pp) | −0.03 | [−0.46, 0.40] | 1 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 8 | Dice | −0.123 | [−0.691, 0.445] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 8 | AJI | 0.022 | [−0.005, 0.048] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 8 | IPE (pp) | 0.44 | [−0.91, 1.78] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 16 | Dice | −0.125 | [−0.710, 0.459] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 16 | AJI | 0.005 | [−0.118, 0.127] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 16 | IPE (pp) | 0.47 | [−1.27, 2.21] | 3 of 3 seeds |



**Table 16:** MobileNet-UNet: paired differences. IPE in percentage points, Dice and AJI as fractions.

| Rank | Comparison (b − a) | Metric | Mean difference | 95% $t$-interval | b higher in |
|---|---|---|---|---|---|
| 2 | +PMC − Dice only | IPE (pp) | 1.90 | [−14.43, 18.22] | 2 of 3 seeds |
| 2 | +PMC − Dice only | Dice | −0.003 | [−0.461, 0.456] | 1 of 3 seeds |
| 2 | +PMC − Dice only | AJI | −0.009 | [−0.127, 0.109] | 1 of 3 seeds |
| 2 | +PMC+BGA − Dice only | IPE (pp) | −5.58 | [−22.19, 11.04] | 1 of 3 seeds |
| 2 | +PMC+BGA − Dice only | Dice | 0.133 | [−0.154, 0.419] | 3 of 3 seeds |
| 2 | +PMC+BGA − Dice only | AJI | 0.068 | [0.043, 0.092] | 3 of 3 seeds |
| 2 | Full (+IPP) − Dice only | IPE (pp) | −15.37 | [−16.63, −14.12] | 0 of 3 seeds |
| 2 | Full (+IPP) − Dice only | Dice | −0.006 | [−0.024, 0.013] | 1 of 3 seeds |
| 2 | Full (+IPP) − Dice only | AJI | −0.045 | [−0.087, −0.004] | 0 of 3 seeds |
| 2 | Full (+IPP) − +PMC+BGA | IPE (pp) | −9.80 | [−25.71, 6.12] | 0 of 3 seeds |
| 2 | Full (+IPP) − +PMC+BGA | Dice | −0.139 | [−0.410, 0.132] | 0 of 3 seeds |
| 2 | Full (+IPP) − +PMC+BGA | AJI | −0.113 | [−0.129, −0.096] | 0 of 3 seeds |
| 4 | +PMC − Dice only | IPE (pp) | 2.87 | [−9.35, 15.08] | 2 of 3 seeds |
| 4 | +PMC − Dice only | Dice | −0.065 | [−0.332, 0.203] | 1 of 3 seeds |
| 4 | +PMC − Dice only | AJI | −0.034 | [−0.255, 0.188] | 2 of 3 seeds |
| 4 | +PMC+BGA − Dice only | IPE (pp) | −1.83 | [−29.53, 25.88] | 2 of 3 seeds |
| 4 | +PMC+BGA − Dice only | Dice | 0.008 | [−0.497, 0.513] | 1 of 3 seeds |
| 4 | +PMC+BGA − Dice only | AJI | −0.068 | [−0.256, 0.120] | 0 of 3 seeds |
| 4 | Full (+IPP) − Dice only | IPE (pp) | −14.02 | [−20.74, −7.31] | 0 of 3 seeds |

| Rank | Comparison (b − a) | Metric | Mean difference | 95% $t$-interval | b higher in |
|---|---|---|---|---|---|
| 4 | Full (+IPP) − Dice only | Dice | −0.075 | [−0.349, 0.198] | 0 of 3 seeds |
| 4 | Full (+IPP) − Dice only | AJI | −0.068 | [−0.179, 0.042] | 0 of 3 seeds |
| 4 | Full (+IPP) − +PMC+BGA | IPE (pp) | −12.20 | [−38.39, 13.99] | 0 of 3 seeds |
| 4 | Full (+IPP) − +PMC+BGA | Dice | −0.083 | [−0.644, 0.478] | 1 of 3 seeds |
| 4 | Full (+IPP) − +PMC+BGA | AJI | −0.000 | [−0.181, 0.180] | 1 of 3 seeds |
| 8 | +PMC − Dice only | IPE (pp) | 4.31 | [−23.66, 32.28] | 2 of 3 seeds |
| 8 | +PMC − Dice only | Dice | −0.071 | [−0.796, 0.654] | 1 of 3 seeds |
| 8 | +PMC − Dice only | AJI | −0.038 | [−0.251, 0.175] | 1 of 3 seeds |
| 8 | +PMC+BGA − Dice only | IPE (pp) | −1.52 | [−9.78, 6.74] | 2 of 3 seeds |
| 8 | +PMC+BGA − Dice only | Dice | 0.059 | [−0.193, 0.311] | 2 of 3 seeds |
| 8 | +PMC+BGA − Dice only | AJI | −0.021 | [−0.109, 0.067] | 1 of 3 seeds |
| 8 | Full (+IPP) − Dice only | IPE (pp) | −12.88 | [−44.66, 18.90] | 1 of 3 seeds |
| 8 | Full (+IPP) − Dice only | Dice | −0.073 | [−0.796, 0.650] | 1 of 3 seeds |
| 8 | Full (+IPP) − Dice only | AJI | −0.073 | [−0.478, 0.332] | 1 of 3 seeds |
| 8 | Full (+IPP) − +PMC+BGA | IPE (pp) | −11.36 | [−42.49, 19.77] | 1 of 3 seeds |
| 8 | Full (+IPP) − +PMC+BGA | Dice | −0.132 | [−0.853, 0.588] | 1 of 3 seeds |
| 8 | Full (+IPP) − +PMC+BGA | AJI | −0.051 | [−0.494, 0.391] | 1 of 3 seeds |
| 16 | +PMC − Dice only | IPE (pp) | 5.95 | [−16.92, 28.82] | 2 of 3 seeds |
| 16 | +PMC − Dice only | Dice | −0.138 | [−0.697, 0.422] | 1 of 3 seeds |
| 16 | +PMC − Dice only | AJI | −0.061 | [−0.223, 0.101] | 1 of 3 seeds |
| 16 | +PMC+BGA − Dice only | IPE (pp) | −2.68 | [−44.52, 39.15] | 1 of 3 seeds |
| 16 | +PMC+BGA − Dice only | Dice | 0.061 | [−0.946, 1.068] | 2 of 3 seeds |
| 16 | +PMC+BGA − Dice only | AJI | 0.022 | [−0.235, 0.280] | 2 of 3 seeds |
| 16 | Full (+IPP) − Dice only | IPE (pp) | −11.09 | [−44.34, 22.17] | 1 of 3 seeds |
| 16 | Full (+IPP) − Dice only | Dice | −0.136 | [−0.871, 0.598] | 1 of 3 seeds |
| 16 | Full (+IPP) − Dice only | AJI | −0.114 | [−0.332, 0.104] | 0 of 3 seeds |
| 16 | Full (+IPP) − +PMC+BGA | IPE (pp) | −8.40 | [−30.79, 13.98] | 1 of 3 seeds |
| 16 | Full (+IPP) − +PMC+BGA | Dice | −0.198 | [−0.691, 0.295] | 0 of 3 seeds |
| 16 | Full (+IPP) − +PMC+BGA | AJI | −0.137 | [−0.250, −0.024] | 0 of 3 seeds |
| 32 | +PMC − Dice only | IPE (pp) | −4.87 | [−17.45, 7.71] | 0 of 3 seeds |
| 32 | +PMC − Dice only | Dice | 0.126 | [−0.137, 0.390] | 3 of 3 seeds |
| 32 | +PMC − Dice only | AJI | 0.007 | [−0.129, 0.142] | 2 of 3 seeds |
| 32 | +PMC+BGA − Dice only | IPE (pp) | −1.29 | [−4.87, 2.30] | 0 of 3 seeds |
| 32 | +PMC+BGA − Dice only | Dice | 0.059 | [−0.204, 0.322] | 1 of 3 seeds |
| 32 | +PMC+BGA − Dice only | AJI | −0.025 | [−0.167, 0.118] | 2 of 3 seeds |
| 32 | Full (+IPP) − Dice only | IPE (pp) | −15.00 | [−40.27, 10.28] | 0 of 3 seeds |
| 32 | Full (+IPP) − Dice only | Dice | 0.117 | [−0.156, 0.390] | 2 of 3 seeds |
| 32 | Full (+IPP) − Dice only | AJI | −0.051 | [−0.178, 0.075] | 1 of 3 seeds |
| 32 | Full (+IPP) − +PMC+BGA | IPE (pp) | −13.71 | [−42.50, 15.08] | 0 of 3 seeds |
| 32 | Full (+IPP) − +PMC+BGA | Dice | 0.058 | [−0.208, 0.324] | 1 of 3 seeds |
| 32 | Full (+IPP) − +PMC+BGA | AJI | −0.027 | [−0.196, 0.142] | 2 of 3 seeds |
| – | Full (+IPP), r = 4 − Full (+IPP), r = 2 | Dice | −0.004 | [−0.018, 0.010] | 1 of 3 seeds |
| – | Full (+IPP), r = 4 − Full (+IPP), r = 2 | AJI | 0.026 | [0.009, 0.044] | 3 of 3 seeds |
| – | Full (+IPP), r = 4 − Full (+IPP), r = 2 | IPE (pp) | −0.11 | [−0.62, 0.41] | 1 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 2 | Dice | 0.001 | [−0.016, 0.019] | 2 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 2 | AJI | 0.016 | [−0.157, 0.190] | 2 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 2 | IPE (pp) | −0.00 | [−0.65, 0.65] | 1 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 2 | Dice | 0.004 | [−0.008, 0.015] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 2 | AJI | −0.029 | [−0.072, 0.014] | 0 of 3 seeds |

| Rank | Comparison (b − a) | Metric | Mean difference | 95% $t$-interval | b higher in |
|---|---|---|---|---|---|
| – | Full (+IPP), r = 16 − Full (+IPP), r = 2 | IPE (pp) | 0.49 | [−1.33, 2.31] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 2 | Dice | 0.066 | [−0.195, 0.326] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 2 | AJI | 0.007 | [−0.175, 0.188] | 1 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 2 | IPE (pp) | 3.23 | [−11.62, 18.07] | 1 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 4 | Dice | 0.005 | [−0.005, 0.016] | 3 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 4 | AJI | −0.010 | [−0.189, 0.170] | 2 of 3 seeds |
| – | Full (+IPP), r = 8 − Full (+IPP), r = 4 | IPE (pp) | 0.10 | [−0.27, 0.48] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 4 | Dice | 0.007 | [0.002, 0.013] | 3 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 4 | AJI | −0.055 | [−0.112, 0.002] | 0 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 4 | IPE (pp) | 0.60 | [−0.71, 1.91] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 4 | Dice | 0.070 | [−0.202, 0.341] | 3 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 4 | AJI | −0.020 | [−0.218, 0.179] | 1 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 4 | IPE (pp) | 3.33 | [−11.18, 17.85] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 8 | Dice | 0.002 | [−0.005, 0.009] | 2 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 8 | AJI | −0.046 | [−0.185, 0.094] | 1 of 3 seeds |
| – | Full (+IPP), r = 16 − Full (+IPP), r = 8 | IPE (pp) | 0.49 | [−0.94, 1.92] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 8 | Dice | 0.065 | [−0.214, 0.343] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 8 | AJI | −0.010 | [−0.211, 0.191] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 8 | IPE (pp) | 3.23 | [−10.99, 17.45] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 16 | Dice | 0.062 | [−0.209, 0.334] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 16 | AJI | 0.036 | [−0.114, 0.185] | 2 of 3 seeds |
| – | Full (+IPP), r = 32 − Full (+IPP), r = 16 | IPE (pp) | 2.74 | [−11.24, 16.71] | 2 of 3 seeds |



# E Every run

**Table 17:** All runs, last epoch. D = Dice of the class, I = IPE of the class in percent, IPE = global IPE in percent.

| Arch | Loss | Rank | Seed | Dice | AJI | BF1 | IPE | D Disco | D Sphero | D Echi | D Stoma | I Disco | I Sphero | I Echi | I Stoma | Min class Dice | Lost classes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| EdgeSAM | Dice only | 2 | 42 | 0.579 | 0.275 | 0.463 | 12.81 | 0.894 | 0.192 | 0.663 | 0.566 | 7.18 | 61.05 | 30.77 | 39.36 | 0.192 | |
| EdgeSAM | Dice only | 2 | 43 | 0.562 | 0.316 | 0.563 | 16.37 | 0.888 | 0.147 | 0.642 | 0.572 | 7.35 | 89.63 | 31.70 | 51.56 | 0.147 | |
| EdgeSAM | Dice only | 2 | 44 | 0.588 | 0.341 | 0.580 | 14.08 | 0.890 | 0.219 | 0.660 | 0.583 | 6.94 | 82.01 | 26.20 | 45.09 | 0.219 | |
| EdgeSAM | Dice only | 4 | 42 | 0.706 | 0.329 | 0.623 | 4.85 | 0.893 | 0.734 | 0.659 | 0.540 | 7.01 | 27.99 | 28.16 | 52.34 | 0.540 | |
| EdgeSAM | Dice only | 4 | 43 | 0.547 | 0.260 | 0.236 | 13.68 | 0.889 | 0.043 | 0.657 | 0.601 | 7.16 | 88.58 | 30.86 | 53.87 | 0.043 | spherocyte |
| EdgeSAM | Dice only | 4 | 44 | 0.576 | 0.374 | 0.570 | 14.34 | 0.887 | 0.215 | 0.631 | 0.572 | 7.35 | 81.84 | 34.21 | 48.87 | 0.215 | |
| EdgeSAM | Dice only | 8 | 42 | 0.730 | 0.348 | 0.643 | 4.60 | 0.896 | 0.761 | 0.677 | 0.585 | 6.45 | 28.73 | 30.63 | 50.39 | 0.585 | |
| EdgeSAM | Dice only | 8 | 43 | 0.546 | 0.276 | 0.256 | 13.93 | 0.892 | 0.045 | 0.652 | 0.596 | 6.57 | 80.92 | 30.67 | 48.38 | 0.045 | spherocyte |
| EdgeSAM | Dice only | 8 | 44 | 0.595 | 0.377 | 0.589 | 13.93 | 0.891 | 0.222 | 0.674 | 0.594 | 6.59 | 83.06 | 24.84 | 49.42 | 0.222 | |
| EdgeSAM | Dice only | 16 | 42 | 0.740 | 0.306 | 0.644 | 4.19 | 0.897 | 0.774 | 0.682 | 0.606 | 6.30 | 20.18 | 29.82 | 49.18 | 0.606 | |
| EdgeSAM | Dice only | 16 | 43 | 0.568 | 0.273 | 0.252 | 13.87 | 0.900 | 0.045 | 0.666 | 0.661 | 5.62 | 80.41 | 31.15 | 44.25 | 0.045 | spherocyte |
| EdgeSAM | Dice only | 16 | 44 | 0.554 | 0.365 | 0.626 | 15.10 | 0.897 | 0.000 | 0.679 | 0.641 | 6.09 | 99.99 | 33.41 | 53.08 | 0.000 | spherocyte |
| EdgeSAM | Dice only | 32 | 42 | 0.744 | 0.328 | 0.649 | 3.75 | 0.899 | 0.779 | 0.674 | 0.625 | 6.75 | 27.91 | 33.40 | 41.33 | 0.625 | |
| EdgeSAM | Dice only | 32 | 43 | 0.565 | 0.314 | 0.274 | 14.72 | 0.896 | 0.038 | 0.680 | 0.644 | 6.77 | 85.60 | 27.06 | 43.62 | 0.038 | spherocyte |
| EdgeSAM | Dice only | 32 | 44 | 0.548 | 0.343 | 0.633 | 15.13 | 0.898 | 0.000 | 0.672 | 0.621 | 7.36 | 99.99 | 32.02 | 45.81 | 0.000 | spherocyte |
| EdgeSAM | +PMC | 2 | 42 | 0.711 | 0.303 | 0.631 | 3.70 | 0.893 | 0.770 | 0.643 | 0.536 | 6.90 | 22.68 | 33.49 | 49.75 | 0.536 | |
| EdgeSAM | +PMC | 2 | 43 | 0.705 | 0.411 | 0.578 | 6.25 | 0.880 | 0.753 | 0.626 | 0.562 | 8.43 | 26.56 | 30.73 | 46.09 | 0.562 | |
| EdgeSAM | +PMC | 2 | 44 | 0.716 | 0.408 | 0.609 | 5.85 | 0.885 | 0.777 | 0.629 | 0.572 | 8.15 | 24.03 | 42.54 | 48.88 | 0.572 | |
| EdgeSAM | +PMC | 4 | 42 | 0.730 | 0.340 | 0.643 | 3.29 | 0.895 | 0.778 | 0.665 | 0.583 | 7.44 | 27.66 | 31.20 | 44.73 | 0.583 | |
| EdgeSAM | +PMC | 4 | 43 | 0.733 | 0.400 | 0.604 | 5.48 | 0.889 | 0.788 | 0.660 | 0.594 | 7.29 | 20.61 | 29.64 | 46.12 | 0.594 | |
| EdgeSAM | +PMC | 4 | 44 | 0.717 | 0.350 | 0.610 | 5.26 | 0.891 | 0.751 | 0.640 | 0.585 | 7.49 | 28.82 | 30.37 | 51.43 | 0.585 | |
| EdgeSAM | +PMC | 8 | 42 | 0.739 | 0.350 | 0.656 | 3.28 | 0.897 | 0.789 | 0.660 | 0.611 | 7.14 | 24.65 | 29.84 | 46.35 | 0.611 | |
| EdgeSAM | +PMC | 8 | 43 | 0.730 | 0.416 | 0.610 | 4.88 | 0.887 | 0.784 | 0.667 | 0.582 | 7.75 | 23.83 | 28.85 | 45.73 | 0.582 | |
| EdgeSAM | +PMC | 8 | 44 | 0.545 | 0.391 | 0.597 | 15.77 | 0.891 | 0.000 | 0.681 | 0.609 | 7.29 | 100.00 | 30.43 | 49.22 | 0.000 | spherocyte |
| EdgeSAM | +PMC | 16 | 42 | 0.751 | 0.400 | 0.653 | 3.47 | 0.897 | 0.783 | 0.686 | 0.637 | 6.47 | 23.19 | 26.32 | 46.15 | 0.637 | |
| EdgeSAM | +PMC | 16 | 43 | 0.739 | 0.320 | 0.637 | 3.15 | 0.896 | 0.783 | 0.687 | 0.592 | 6.19 | 25.45 | 26.56 | 48.79 | 0.592 | |

| Arch | Loss | Rank | Seed | Dice | AJI | BF1 | IPE | D Disco | D Sphero | D Echi | D Stoma | I Disco | I Sphero | I Echi | I Stoma | Min class Dice | Lost classes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| EdgeSAM | +PMC | 16 | 44 | 0.731 | 0.474 | 0.638 | 4.52 | 0.890 | 0.781 | 0.670 | 0.584 | 7.85 | 20.48 | 34.97 | 48.12 | 0.584 | |
| EdgeSAM | +PMC | 32 | 42 | 0.747 | 0.329 | 0.659 | 3.25 | 0.898 | 0.779 | 0.680 | 0.631 | 6.30 | 19.98 | 27.73 | 43.32 | 0.631 | |
| EdgeSAM | +PMC | 32 | 43 | 0.744 | 0.388 | 0.647 | 3.38 | 0.897 | 0.785 | 0.674 | 0.618 | 6.56 | 25.09 | 29.86 | 48.29 | 0.618 | |
| EdgeSAM | +PMC | 32 | 44 | 0.550 | 0.448 | 0.607 | 15.74 | 0.892 | 0.000 | 0.679 | 0.628 | 6.63 | 100.00 | 26.29 | 51.81 | 0.000 | spherocyte |
| EdgeSAM | +PMC+BGA | 2 | 42 | 0.718 | 0.325 | 0.635 | 3.28 | 0.891 | 0.790 | 0.639 | 0.551 | 7.83 | 22.88 | 32.93 | 46.58 | 0.551 | |
| EdgeSAM | +PMC+BGA | 2 | 43 | 0.723 | 0.397 | 0.600 | 4.91 | 0.887 | 0.781 | 0.634 | 0.591 | 7.97 | 26.88 | 34.12 | 47.64 | 0.591 | |
| EdgeSAM | +PMC+BGA | 2 | 44 | 0.711 | 0.433 | 0.595 | 6.45 | 0.885 | 0.760 | 0.632 | 0.566 | 8.30 | 28.05 | 31.89 | 56.71 | 0.566 | |
| EdgeSAM | +PMC+BGA | 4 | 42 | 0.728 | 0.348 | 0.642 | 3.33 | 0.896 | 0.779 | 0.639 | 0.599 | 7.41 | 19.47 | 32.44 | 41.77 | 0.599 | |
| EdgeSAM | +PMC+BGA | 4 | 43 | 0.731 | 0.405 | 0.602 | 5.14 | 0.887 | 0.767 | 0.655 | 0.614 | 7.76 | 19.44 | 30.68 | 41.18 | 0.614 | |
| EdgeSAM | +PMC+BGA | 4 | 44 | 0.711 | 0.423 | 0.597 | 6.46 | 0.886 | 0.763 | 0.628 | 0.568 | 7.35 | 26.35 | 35.24 | 53.32 | 0.568 | |
| EdgeSAM | +PMC+BGA | 8 | 42 | 0.747 | 0.341 | 0.642 | 3.01 | 0.896 | 0.796 | 0.660 | 0.635 | 6.82 | 21.43 | 32.31 | 43.54 | 0.635 | |
| EdgeSAM | +PMC+BGA | 8 | 43 | 0.731 | 0.400 | 0.615 | 4.82 | 0.889 | 0.771 | 0.664 | 0.601 | 8.04 | 21.43 | 29.22 | 43.05 | 0.601 | |
| EdgeSAM | +PMC+BGA | 8 | 44 | 0.728 | 0.431 | 0.628 | 4.91 | 0.890 | 0.783 | 0.637 | 0.600 | 8.08 | 23.50 | 33.05 | 53.37 | 0.600 | |
| EdgeSAM | +PMC+BGA | 16 | 42 | 0.748 | 0.312 | 0.654 | 2.73 | 0.899 | 0.789 | 0.682 | 0.623 | 5.89 | 23.78 | 27.68 | 46.16 | 0.623 | |
| EdgeSAM | +PMC+BGA | 16 | 43 | 0.740 | 0.354 | 0.638 | 3.25 | 0.895 | 0.767 | 0.681 | 0.616 | 6.63 | 30.88 | 25.31 | 47.41 | 0.616 | |
| EdgeSAM | +PMC+BGA | 16 | 44 | 0.738 | 0.352 | 0.639 | 4.32 | 0.894 | 0.777 | 0.665 | 0.613 | 6.25 | 22.72 | 28.07 | 53.60 | 0.613 | |
| EdgeSAM | +PMC+BGA | 32 | 42 | 0.753 | 0.369 | 0.642 | 3.34 | 0.897 | 0.777 | 0.684 | 0.654 | 5.30 | 24.18 | 31.12 | 44.92 | 0.654 | |
| EdgeSAM | +PMC+BGA | 32 | 43 | 0.741 | 0.411 | 0.655 | 3.02 | 0.898 | 0.781 | 0.675 | 0.611 | 6.36 | 29.62 | 29.63 | 41.24 | 0.611 | |
| EdgeSAM | +PMC+BGA | 32 | 44 | 0.743 | 0.437 | 0.633 | 4.57 | 0.893 | 0.786 | 0.676 | 0.615 | 7.40 | 20.75 | 30.51 | 47.76 | 0.615 | |
| EdgeSAM | Full (+IPP) | 2 | 42 | 0.685 | 0.257 | 0.621 | 2.86 | 0.891 | 0.733 | 0.620 | 0.494 | 6.69 | 25.44 | 35.80 | 67.52 | 0.494 | |
| EdgeSAM | Full (+IPP) | 2 | 43 | 0.691 | 0.230 | 0.592 | 2.93 | 0.882 | 0.758 | 0.605 | 0.520 | 7.68 | 28.17 | 34.66 | 54.84 | 0.520 | |
| EdgeSAM | Full (+IPP) | 2 | 44 | 0.501 | 0.260 | 0.582 | 7.58 | 0.874 | 0.000 | 0.623 | 0.506 | 25.74 | 100.00 | 39.87 | 58.61 | 0.000 | spherocyte |
| EdgeSAM | Full (+IPP) | 4 | 42 | 0.710 | 0.233 | 0.627 | 2.99 | 0.894 | 0.779 | 0.620 | 0.547 | 6.60 | 31.14 | 33.71 | 57.12 | 0.547 | |
| EdgeSAM | Full (+IPP) | 4 | 43 | 0.707 | 0.249 | 0.605 | 2.91 | 0.889 | 0.767 | 0.606 | 0.567 | 6.12 | 29.92 | 35.64 | 45.36 | 0.567 | |
| EdgeSAM | Full (+IPP) | 4 | 44 | 0.693 | 0.275 | 0.611 | 4.53 | 0.891 | 0.754 | 0.611 | 0.517 | 6.65 | 30.05 | 37.30 | 48.79 | 0.517 | |
| EdgeSAM | Full (+IPP) | 8 | 42 | 0.706 | 0.224 | 0.615 | 3.13 | 0.891 | 0.765 | 0.621 | 0.545 | 6.41 | 29.41 | 32.03 | 53.97 | 0.545 | |
| EdgeSAM | Full (+IPP) | 8 | 43 | 0.719 | 0.225 | 0.611 | 2.43 | 0.887 | 0.774 | 0.632 | 0.583 | 7.00 | 25.15 | 33.59 | 41.17 | 0.583 | |
| EdgeSAM | Full (+IPP) | 8 | 44 | 0.515 | 0.299 | 0.598 | 5.87 | 0.869 | 0.000 | 0.632 | 0.561 | 54.06 | 100.00 | 37.15 | 53.59 | 0.000 | spherocyte |
| EdgeSAM | Full (+IPP) | 16 | 42 | 0.722 | 0.274 | 0.636 | 2.90 | 0.894 | 0.777 | 0.644 | 0.573 | 6.17 | 24.04 | 35.17 | 58.38 | 0.573 | |
| EdgeSAM | Full (+IPP) | 16 | 43 | 0.740 | 0.257 | 0.639 | 2.37 | 0.893 | 0.786 | 0.657 | 0.622 | 6.62 | 23.93 | 28.28 | 47.29 | 0.622 | |
| EdgeSAM | Full (+IPP) | 16 | 44 | 0.533 | 0.301 | 0.596 | 8.73 | 0.881 | 0.000 | 0.674 | 0.578 | 29.63 | 100.00 | 33.10 | 61.93 | 0.000 | spherocyte |
| EdgeSAM | Full (+IPP) | 32 | 42 | 0.731 | 0.329 | 0.647 | 2.66 | 0.896 | 0.784 | 0.652 | 0.593 | 6.19 | 28.98 | 29.42 | 53.53 | 0.593 | |
| EdgeSAM | Full (+IPP) | 32 | 43 | 0.743 | 0.320 | 0.642 | 2.41 | 0.894 | 0.791 | 0.666 | 0.619 | 6.80 | 20.83 | 28.68 | 54.00 | 0.619 | |
| EdgeSAM | Full (+IPP) | 32 | 44 | 0.740 | 0.358 | 0.638 | 3.98 | 0.894 | 0.793 | 0.645 | 0.629 | 6.34 | 22.93 | 28.12 | 49.09 | 0.629 | |
| MobileSAM | Dice only | 2 | 42 | 0.599 | 0.200 | 0.740 | 13.01 | 0.923 | 0.000 | 0.772 | 0.702 | 7.48 | 100.00 | 20.46 | 48.62 | 0.000 | spherocyte |
| MobileSAM | Dice only | 2 | 43 | 0.811 | 0.260 | 0.805 | 1.44 | 0.923 | 0.852 | 0.753 | 0.717 | 6.96 | 23.73 | 30.94 | 49.19 | 0.717 | |
| MobileSAM | Dice only | 2 | 44 | 0.812 | 0.195 | 0.794 | 1.62 | 0.924 | 0.842 | 0.772 | 0.713 | 6.17 | 25.27 | 23.03 | 45.21 | 0.713 | |
| MobileSAM | Dice only | 4 | 42 | 0.812 | 0.239 | 0.809 | 1.58 | 0.925 | 0.842 | 0.759 | 0.722 | 6.80 | 25.81 | 24.89 | 50.34 | 0.722 | |
| MobileSAM | Dice only | 4 | 43 | 0.809 | 0.280 | 0.798 | 1.28 | 0.924 | 0.853 | 0.753 | 0.707 | 6.16 | 20.30 | 30.45 | 48.82 | 0.707 | |
| MobileSAM | Dice only | 4 | 44 | 0.813 | 0.284 | 0.797 | 1.46 | 0.922 | 0.851 | 0.770 | 0.708 | 7.71 | 21.99 | 24.87 | 52.19 | 0.708 | |
| MobileSAM | Dice only | 8 | 42 | 0.805 | 0.222 | 0.799 | 1.81 | 0.921 | 0.844 | 0.748 | 0.708 | 8.01 | 20.60 | 26.03 | 53.80 | 0.708 | |
| MobileSAM | Dice only | 8 | 43 | 0.813 | 0.260 | 0.803 | 1.27 | 0.923 | 0.850 | 0.756 | 0.721 | 5.98 | 16.85 | 26.82 | 44.42 | 0.721 | |
| MobileSAM | Dice only | 8 | 44 | 0.815 | 0.261 | 0.821 | 1.51 | 0.927 | 0.854 | 0.769 | 0.710 | 6.38 | 25.02 | 28.55 | 47.11 | 0.710 | |
| MobileSAM | Dice only | 16 | 42 | 0.812 | 0.218 | 0.795 | 1.68 | 0.923 | 0.851 | 0.760 | 0.715 | 6.87 | 22.68 | 30.23 | 46.08 | 0.715 | |
| MobileSAM | Dice only | 16 | 43 | 0.816 | 0.225 | 0.806 | 1.45 | 0.926 | 0.851 | 0.786 | 0.700 | 6.40 | 20.97 | 28.01 | 47.66 | 0.700 | |
| MobileSAM | Dice only | 16 | 44 | 0.812 | 0.235 | 0.805 | 1.50 | 0.924 | 0.853 | 0.752 | 0.717 | 5.83 | 21.04 | 30.60 | 45.68 | 0.717 | |
| MobileSAM | Dice only | 32 | 42 | 0.602 | 0.236 | 0.750 | 12.88 | 0.924 | 0.000 | 0.767 | 0.716 | 6.24 | 100.00 | 23.48 | 47.51 | 0.000 | spherocyte |
| MobileSAM | Dice only | 32 | 43 | 0.600 | 0.189 | 0.740 | 13.24 | 0.923 | 0.000 | 0.757 | 0.720 | 6.75 | 100.00 | 27.99 | 51.46 | 0.000 | spherocyte |
| MobileSAM | Dice only | 32 | 44 | 0.819 | 0.263 | 0.819 | 1.58 | 0.926 | 0.853 | 0.766 | 0.732 | 6.07 | 23.58 | 29.04 | 52.85 | 0.732 | |
| MobileSAM | +PMC | 2 | 42 | 0.801 | 0.201 | 0.798 | 1.67 | 0.921 | 0.839 | 0.736 | 0.709 | 7.25 | 21.06 | 24.78 | 50.02 | 0.709 | |
| MobileSAM | +PMC | 2 | 43 | 0.813 | 0.232 | 0.799 | 1.43 | 0.924 | 0.849 | 0.762 | 0.717 | 7.23 | 26.95 | 23.51 | 53.95 | 0.717 | |
| MobileSAM | +PMC | 2 | 44 | 0.800 | 0.252 | 0.803 | 1.57 | 0.923 | 0.850 | 0.760 | 0.666 | 6.96 | 27.17 | 26.91 | 42.72 | 0.666 | |
| MobileSAM | +PMC | 4 | 42 | 0.800 | 0.212 | 0.801 | 1.64 | 0.923 | 0.840 | 0.735 | 0.702 | 7.30 | 22.82 | 24.78 | 45.66 | 0.702 | |
| MobileSAM | +PMC | 4 | 43 | 0.812 | 0.262 | 0.793 | 1.38 | 0.923 | 0.851 | 0.763 | 0.709 | 7.64 | 17.80 | 28.41 | 50.69 | 0.709 | |
| MobileSAM | +PMC | 4 | 44 | 0.805 | 0.233 | 0.803 | 1.46 | 0.922 | 0.858 | 0.766 | 0.674 | 6.79 | 29.14 | 25.03 | 51.46 | 0.674 | |
| MobileSAM | +PMC | 8 | 42 | 0.813 | 0.202 | 0.806 | 1.86 | 0.924 | 0.846 | 0.767 | 0.715 | 6.76 | 25.34 | 26.22 | 47.06 | 0.715 | |
| MobileSAM | +PMC | 8 | 43 | 0.822 | 0.253 | 0.803 | 1.38 | 0.927 | 0.859 | 0.784 | 0.719 | 6.01 | 21.74 | 24.58 | 53.47 | 0.719 | |
| MobileSAM | +PMC | 8 | 44 | 0.804 | 0.270 | 0.807 | 1.56 | 0.922 | 0.851 | 0.773 | 0.670 | 6.77 | 24.15 | 25.27 | 50.93 | 0.670 | |
| MobileSAM | +PMC | 16 | 42 | 0.808 | 0.211 | 0.805 | 1.68 | 0.924 | 0.838 | 0.757 | 0.712 | 6.99 | 17.97 | 23.33 | 46.35 | 0.712 | |
| MobileSAM | +PMC | 16 | 43 | 0.821 | 0.280 | 0.809 | 1.31 | 0.927 | 0.861 | 0.770 | 0.725 | 5.75 | 20.10 | 24.97 | 50.15 | 0.725 | |
| MobileSAM | +PMC | 16 | 44 | 0.811 | 0.240 | 0.818 | 1.75 | 0.926 | 0.845 | 0.768 | 0.705 | 7.25 | 26.90 | 25.41 | 55.47 | 0.705 | |
| MobileSAM | +PMC | 32 | 42 | 0.820 | 0.268 | 0.807 | 1.63 | 0.924 | 0.858 | 0.766 | 0.731 | 7.20 | 18.09 | 24.68 | 52.61 | 0.731 | |
| MobileSAM | +PMC | 32 | 43 | 0.818 | 0.279 | 0.812 | 1.35 | 0.927 | 0.854 | 0.771 | 0.722 | 5.31 | 18.09 | 23.80 | 48.37 | 0.722 | |
| MobileSAM | +PMC | 32 | 44 | 0.812 | 0.246 | 0.821 | 1.75 | 0.925 | 0.865 | 0.772 | 0.686 | 6.91 | 23.21 | 25.45 | 56.18 | 0.686 | |
| MobileSAM | +PMC+BGA | 2 | 42 | 0.596 | 0.215 | 0.734 | 12.93 | 0.921 | 0.000 | 0.759 | 0.705 | 6.97 | 100.00 | 24.47 | 49.99 | 0.000 | spherocyte |
| MobileSAM | +PMC+BGA | 2 | 43 | 0.812 | 0.265 | 0.799 | 1.46 | 0.924 | 0.858 | 0.771 | 0.697 | 6.95 | 24.90 | 22.13 | 53.61 | 0.697 | |
| MobileSAM | +PMC+BGA | 2 | 44 | 0.813 | 0.232 | 0.795 | 1.56 | 0.922 | 0.857 | 0.779 | 0.693 | 5.98 | 21.49 | 22.57 | 43.43 | 0.693 | |
| MobileSAM | +PMC+BGA | 4 | 42 | 0.599 | 0.204 | 0.741 | 12.83 | 0.923 | 0.000 | 0.766 | 0.709 | 7.12 | 100.00 | 26.28 | 47.64 | 0.000 | spherocyte |
| MobileSAM | +PMC+BGA | 4 | 43 | 0.812 | 0.267 | 0.803 | 1.28 | 0.925 | 0.846 | 0.760 | 0.716 | 6.08 | 25.42 | 34.16 | 53.28 | 0.716 | |
| MobileSAM | +PMC+BGA | 4 | 44 | 0.811 | 0.269 | 0.804 | 1.60 | 0.925 | 0.846 | 0.764 | 0.708 | 6.64 | 22.06 | 26.91 | 41.47 | 0.708 | |
| MobileSAM | +PMC+BGA | 8 | 42 | 0.811 | 0.224 | 0.806 | 1.58 | 0.924 | 0.851 | 0.766 | 0.704 | 6.88 | 19.34 | 25.34 | 52.60 | 0.704 | |
| MobileSAM | +PMC+BGA | 8 | 43 | 0.819 | 0.260 | 0.802 | 1.36 | 0.925 | 0.859 | 0.763 | 0.730 | 7.70 | 23.71 | 26.73 | 50.84 | 0.730 | |
| MobileSAM | +PMC+BGA | 8 | 44 | 0.810 | 0.293 | 0.810 | 1.50 | 0.924 | 0.853 | 0.769 | 0.695 | 6.44 | 17.89 | 27.24 | 58.36 | 0.695 | |
| MobileSAM | +PMC+BGA | 16 | 42 | 0.815 | 0.234 | 0.802 | 1.68 | 0.925 | 0.848 | 0.765 | 0.723 | 6.31 | 20.49 | 28.68 | 48.05 | 0.723 | |
| MobileSAM | +PMC+BGA | 16 | 43 | 0.820 | 0.224 | 0.804 | 1.50 | 0.927 | 0.854 | 0.771 | 0.730 | 5.62 | 27.42 | 28.66 | 54.18 | 0.730 | |
| MobileSAM | +PMC+BGA | 16 | 44 | 0.815 | 0.253 | 0.816 | 1.46 | 0.925 | 0.856 | 0.774 | 0.704 | 7.12 | 24.39 | 25.39 | 47.30 | 0.704 | |
| MobileSAM | +PMC+BGA | 32 | 42 | 0.812 | 0.252 | 0.817 | 1.63 | 0.925 | 0.857 | 0.761 | 0.707 | 7.69 | 19.20 | 26.14 | 42.75 | 0.707 | |
| MobileSAM | +PMC+BGA | 32 | 43 | 0.825 | 0.258 | 0.813 | 1.32 | 0.926 | 0.865 | 0.785 | 0.726 | 6.66 | 19.72 | 27.32 | 49.59 | 0.726 | |

| Arch | Loss | Rank | Seed | Dice | AJI | BF1 | IPE | D Disco | D Sphero | D Echi | D Stoma | I Disco | I Sphero | I Echi | I Stoma | Min class Dice | Lost classes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MobileSAM | +PMC+BGA | 32 | 44 | 0.815 | 0.287 | 0.811 | 1.44 | 0.925 | 0.853 | 0.775 | 0.708 | 6.75 | 21.23 | 25.82 | 47.48 | 0.708 | |
| MobileSAM | Full (+IPP) | 2 | 42 | 0.806 | 0.261 | 0.801 | 1.38 | 0.923 | 0.849 | 0.766 | 0.687 | 6.48 | 27.73 | 24.51 | 56.51 | 0.687 | |
| MobileSAM | Full (+IPP) | 2 | 43 | 0.807 | 0.278 | 0.795 | 1.21 | 0.924 | 0.850 | 0.767 | 0.685 | 7.30 | 21.73 | 24.54 | 42.99 | 0.685 | |
| MobileSAM | Full (+IPP) | 2 | 44 | 0.812 | 0.253 | 0.808 | 1.31 | 0.924 | 0.847 | 0.768 | 0.709 | 6.94 | 21.98 | 23.95 | 51.18 | 0.709 | |
| MobileSAM | Full (+IPP) | 4 | 42 | 0.807 | 0.227 | 0.803 | 1.42 | 0.924 | 0.854 | 0.773 | 0.678 | 6.93 | 20.14 | 24.31 | 51.77 | 0.678 | |
| MobileSAM | Full (+IPP) | 4 | 43 | 0.815 | 0.298 | 0.802 | 1.28 | 0.925 | 0.851 | 0.770 | 0.715 | 6.99 | 19.68 | 24.70 | 47.79 | 0.715 | |
| MobileSAM | Full (+IPP) | 4 | 44 | 0.809 | 0.280 | 0.805 | 1.33 | 0.921 | 0.861 | 0.775 | 0.680 | 7.47 | 18.32 | 24.35 | 55.43 | 0.680 | |
| MobileSAM | Full (+IPP) | 8 | 42 | 0.810 | 0.243 | 0.792 | 1.42 | 0.921 | 0.855 | 0.763 | 0.700 | 7.60 | 23.15 | 29.93 | 58.36 | 0.700 | |
| MobileSAM | Full (+IPP) | 8 | 43 | 0.805 | 0.274 | 0.801 | 1.30 | 0.921 | 0.850 | 0.754 | 0.693 | 7.42 | 22.34 | 32.26 | 55.46 | 0.693 | |
| MobileSAM | Full (+IPP) | 8 | 44 | 0.817 | 0.268 | 0.813 | 1.24 | 0.926 | 0.847 | 0.764 | 0.730 | 6.44 | 20.88 | 24.06 | 51.64 | 0.730 | |
| MobileSAM | Full (+IPP) | 16 | 42 | 0.811 | 0.221 | 0.803 | 1.41 | 0.922 | 0.855 | 0.778 | 0.687 | 6.58 | 22.12 | 26.36 | 54.53 | 0.687 | |
| MobileSAM | Full (+IPP) | 16 | 43 | 0.814 | 0.287 | 0.787 | 1.08 | 0.921 | 0.856 | 0.783 | 0.697 | 6.26 | 23.28 | 29.45 | 54.50 | 0.697 | |
| MobileSAM | Full (+IPP) | 16 | 44 | 0.813 | 0.328 | 0.813 | 1.37 | 0.926 | 0.852 | 0.770 | 0.705 | 6.69 | 20.38 | 24.05 | 49.18 | 0.705 | |
| MobileSAM | Full (+IPP) | 32 | 42 | 0.818 | 0.266 | 0.799 | 1.51 | 0.923 | 0.857 | 0.772 | 0.718 | 6.53 | 18.12 | 27.74 | 48.17 | 0.718 | |
| MobileSAM | Full (+IPP) | 32 | 43 | 0.417 | 0.306 | 0.768 | 2.36 | 0.890 | 0.000 | 0.779 | 0.000 | 70.74 | 100.00 | 32.32 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileSAM | Full (+IPP) | 32 | 44 | 0.827 | 0.278 | 0.815 | 1.40 | 0.927 | 0.863 | 0.780 | 0.738 | 5.80 | 21.60 | 23.84 | 48.47 | 0.738 | |
| MobileNet-UNet | Dice only | 2 | 42 | 0.420 | 0.565 | 0.794 | 17.54 | 0.928 | 0.000 | 0.753 | 0.000 | 7.25 | 100.00 | 33.34 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Dice only | 2 | 43 | 0.415 | 0.610 | 0.789 | 18.29 | 0.925 | 0.000 | 0.737 | 0.000 | 6.84 | 100.00 | 32.00 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Dice only | 2 | 44 | 0.413 | 0.563 | 0.786 | 18.11 | 0.925 | 0.000 | 0.727 | 0.000 | 6.86 | 100.00 | 27.58 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Dice only | 4 | 42 | 0.420 | 0.645 | 0.795 | 17.40 | 0.926 | 0.000 | 0.756 | 0.000 | 6.58 | 100.00 | 30.44 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Dice only | 4 | 43 | 0.417 | 0.610 | 0.788 | 18.81 | 0.927 | 0.000 | 0.743 | 0.000 | 6.97 | 100.00 | 28.06 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Dice only | 4 | 44 | 0.608 | 0.630 | 0.814 | 13.36 | 0.930 | 0.000 | 0.781 | 0.721 | 6.27 | 100.00 | 25.27 | 50.72 | 0.000 | spherocyte |
| MobileNet-UNet | Dice only | 8 | 42 | 0.232 | 0.534 | 0.775 | 26.39 | 0.930 | 0.000 | 0.000 | 0.000 | 7.23 | 100.00 | 100.00 | 100.00 | 0.000 | spherocyte echinocyte stomatocyte |
| MobileNet-UNet | Dice only | 8 | 43 | 0.808 | 0.748 | 0.875 | 1.49 | 0.929 | 0.847 | 0.754 | 0.699 | 5.16 | 18.70 | 28.21 | 50.99 | 0.699 | |
| MobileNet-UNet | Dice only | 8 | 44 | 0.415 | 0.589 | 0.790 | 18.55 | 0.926 | 0.000 | 0.733 | 0.000 | 7.13 | 100.00 | 29.38 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Dice only | 16 | 42 | 0.232 | 0.532 | 0.760 | 27.11 | 0.929 | 0.000 | 0.000 | 0.000 | 7.31 | 100.00 | 100.00 | 100.00 | 0.000 | spherocyte echinocyte stomatocyte |
| MobileNet-UNet | Dice only | 16 | 43 | 0.819 | 0.710 | 0.867 | 1.61 | 0.931 | 0.850 | 0.754 | 0.742 | 5.21 | 22.75 | 28.19 | 44.79 | 0.742 | |
| MobileNet-UNet | Dice only | 16 | 44 | 0.600 | 0.614 | 0.817 | 13.83 | 0.931 | 0.000 | 0.739 | 0.730 | 5.87 | 100.00 | 26.87 | 46.25 | 0.000 | spherocyte |
| MobileNet-UNet | Dice only | 32 | 42 | 0.232 | 0.572 | 0.760 | 26.88 | 0.929 | 0.000 | 0.000 | 0.000 | 7.17 | 100.00 | 100.00 | 100.00 | 0.000 | spherocyte echinocyte stomatocyte |
| MobileNet-UNet | Dice only | 32 | 43 | 0.424 | 0.617 | 0.793 | 18.99 | 0.929 | 0.000 | 0.765 | 0.000 | 5.25 | 100.00 | 26.21 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Dice only | 32 | 44 | 0.421 | 0.588 | 0.795 | 16.62 | 0.925 | 0.000 | 0.760 | 0.000 | 6.16 | 100.00 | 29.06 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | +PMC | 2 | 42 | 0.232 | 0.513 | 0.751 | 26.48 | 0.926 | 0.000 | 0.000 | 0.000 | 7.66 | 100.00 | 100.00 | 100.00 | 0.000 | spherocyte echinocyte stomatocyte |
| MobileNet-UNet | +PMC | 2 | 43 | 0.415 | 0.593 | 0.781 | 19.10 | 0.926 | 0.000 | 0.735 | 0.000 | 6.35 | 100.00 | 33.10 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | +PMC | 2 | 44 | 0.594 | 0.605 | 0.812 | 14.05 | 0.928 | 0.000 | 0.749 | 0.698 | 5.75 | 100.00 | 31.48 | 53.17 | 0.000 | spherocyte |
| MobileNet-UNet | +PMC | 4 | 42 | 0.231 | 0.509 | 0.749 | 25.94 | 0.926 | 0.000 | 0.000 | 0.000 | 8.57 | 100.00 | 100.00 | 100.00 | 0.000 | spherocyte echinocyte stomatocyte |
| MobileNet-UNet | +PMC | 4 | 43 | 0.418 | 0.636 | 0.783 | 18.61 | 0.927 | 0.000 | 0.746 | 0.000 | 5.95 | 100.00 | 35.66 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | +PMC | 4 | 44 | 0.602 | 0.641 | 0.813 | 13.62 | 0.928 | 0.000 | 0.760 | 0.721 | 4.88 | 100.00 | 26.47 | 50.37 | 0.000 | spherocyte |
| MobileNet-UNet | +PMC | 8 | 42 | 0.232 | 0.503 | 0.758 | 27.08 | 0.927 | 0.000 | 0.000 | 0.000 | 7.52 | 100.00 | 100.00 | 100.00 | 0.000 | spherocyte echinocyte stomatocyte |
| MobileNet-UNet | +PMC | 8 | 43 | 0.416 | 0.621 | 0.789 | 18.43 | 0.927 | 0.000 | 0.736 | 0.000 | 6.52 | 100.00 | 31.82 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | +PMC | 8 | 44 | 0.594 | 0.632 | 0.800 | 13.87 | 0.923 | 0.000 | 0.764 | 0.688 | 6.62 | 100.00 | 31.09 | 47.32 | 0.000 | spherocyte |
| MobileNet-UNet | +PMC | 16 | 42 | 0.232 | 0.445 | 0.760 | 26.64 | 0.929 | 0.000 | 0.000 | 0.000 | 6.87 | 100.00 | 100.00 | 100.00 | 0.000 | spherocyte echinocyte stomatocyte |
| MobileNet-UNet | +PMC | 16 | 43 | 0.422 | 0.601 | 0.793 | 18.11 | 0.928 | 0.000 | 0.759 | 0.000 | 5.65 | 100.00 | 32.17 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | +PMC | 16 | 44 | 0.585 | 0.627 | 0.791 | 15.64 | 0.925 | 0.000 | 0.723 | 0.691 | 6.71 | 100.00 | 31.05 | 40.84 | 0.000 | spherocyte |
| MobileNet-UNet | +PMC | 32 | 42 | 0.426 | 0.617 | 0.801 | 16.30 | 0.928 | 0.000 | 0.775 | 0.000 | 7.22 | 100.00 | 24.35 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | +PMC | 32 | 43 | 0.428 | 0.561 | 0.788 | 18.06 | 0.927 | 0.000 | 0.784 | 0.000 | 6.00 | 100.00 | 24.31 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | +PMC | 32 | 44 | 0.603 | 0.618 | 0.813 | 13.52 | 0.928 | 0.000 | 0.755 | 0.730 | 5.08 | 100.00 | 33.24 | 49.72 | 0.000 | spherocyte |
| MobileNet-UNet | +PMC+BGA | 2 | 42 | 0.422 | 0.643 | 0.788 | 17.75 | 0.927 | 0.000 | 0.759 | 0.000 | 6.57 | 100.00 | 27.29 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | +PMC+BGA | 2 | 43 | 0.633 | 0.676 | 0.819 | 5.39 | 0.924 | 0.861 | 0.745 | 0.000 | 6.50 | 17.68 | 35.82 | 100.00 | 0.000 | stomatocyte |
| MobileNet-UNet | +PMC+BGA | 2 | 44 | 0.593 | 0.622 | 0.803 | 14.06 | 0.925 | 0.000 | 0.751 | 0.697 | 6.24 | 100.00 | 29.65 | 49.82 | 0.000 | spherocyte |
| MobileNet-UNet | +PMC+BGA | 4 | 42 | 0.231 | 0.492 | 0.740 | 25.67 | 0.925 | 0.000 | 0.000 | 0.000 | 8.29 | 100.00 | 100.00 | 100.00 | 0.000 | spherocyte echinocyte stomatocyte |
| MobileNet-UNet | +PMC+BGA | 4 | 43 | 0.635 | 0.602 | 0.819 | 5.02 | 0.923 | 0.858 | 0.757 | 0.000 | 6.30 | 19.42 | 30.47 | 100.00 | 0.000 | stomatocyte |
| MobileNet-UNet | +PMC+BGA | 4 | 44 | 0.604 | 0.588 | 0.811 | 13.40 | 0.929 | 0.000 | 0.759 | 0.726 | 5.78 | 100.00 | 28.72 | 47.40 | 0.000 | spherocyte |
| MobileNet-UNet | +PMC+BGA | 8 | 42 | 0.232 | 0.477 | 0.756 | 26.79 | 0.928 | 0.000 | 0.000 | 0.000 | 7.18 | 100.00 | 100.00 | 100.00 | 0.000 | spherocyte echinocyte stomatocyte |
| MobileNet-UNet | +PMC+BGA | 8 | 43 | 0.810 | 0.727 | 0.857 | 1.90 | 0.925 | 0.851 | 0.754 | 0.708 | 5.87 | 19.64 | 31.18 | 55.24 | 0.708 | |
| MobileNet-UNet | +PMC+BGA | 8 | 44 | 0.591 | 0.603 | 0.825 | 13.19 | 0.927 | 0.000 | 0.737 | 0.701 | 6.08 | 100.00 | 38.35 | 48.52 | 0.000 | spherocyte |
| MobileNet-UNet | +PMC+BGA | 16 | 42 | 0.600 | 0.631 | 0.791 | 14.76 | 0.925 | 0.000 | 0.760 | 0.715 | 6.90 | 100.00 | 22.76 | 50.12 | 0.000 | spherocyte |
| MobileNet-UNet | +PMC+BGA | 16 | 43 | 0.421 | 0.615 | 0.788 | 18.37 | 0.927 | 0.000 | 0.756 | 0.000 | 6.36 | 100.00 | 29.47 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | +PMC+BGA | 16 | 44 | 0.814 | 0.679 | 0.871 | 1.37 | 0.930 | 0.852 | 0.746 | 0.729 | 6.77 | 20.53 | 29.07 | 43.34 | 0.729 | |
| MobileNet-UNet | +PMC+BGA | 32 | 42 | 0.232 | 0.481 | 0.760 | 26.67 | 0.928 | 0.000 | 0.000 | 0.000 | 7.24 | 100.00 | 100.00 | 100.00 | 0.000 | spherocyte echinocyte stomatocyte |
| MobileNet-UNet | +PMC+BGA | 32 | 43 | 0.420 | 0.633 | 0.791 | 18.27 | 0.927 | 0.000 | 0.753 | 0.000 | 6.62 | 100.00 | 26.17 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | +PMC+BGA | 32 | 44 | 0.603 | 0.588 | 0.814 | 13.70 | 0.929 | 0.000 | 0.750 | 0.731 | 6.14 | 100.00 | 27.38 | 42.97 | 0.000 | spherocyte |
| MobileNet-UNet | Full (+IPP) | 2 | 42 | 0.406 | 0.538 | 0.809 | 2.73 | 0.889 | 0.000 | 0.736 | 0.000 | 84.24 | 100.00 | 34.09 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 2 | 43 | 0.411 | 0.561 | 0.820 | 2.75 | 0.888 | 0.000 | 0.756 | 0.000 | 78.87 | 100.00 | 29.81 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 2 | 44 | 0.414 | 0.504 | 0.820 | 2.34 | 0.890 | 0.000 | 0.767 | 0.000 | 82.77 | 100.00 | 27.83 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 4 | 42 | 0.408 | 0.569 | 0.801 | 2.39 | 0.889 | 0.000 | 0.743 | 0.000 | 84.47 | 100.00 | 31.77 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 4 | 43 | 0.406 | 0.590 | 0.802 | 2.72 | 0.889 | 0.000 | 0.735 | 0.000 | 82.62 | 100.00 | 34.25 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 4 | 44 | 0.405 | 0.522 | 0.812 | 2.39 | 0.887 | 0.000 | 0.734 | 0.000 | 80.37 | 100.00 | 33.37 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 8 | 42 | 0.412 | 0.603 | 0.816 | 2.52 | 0.891 | 0.000 | 0.757 | 0.000 | 79.25 | 100.00 | 31.46 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 8 | 43 | 0.416 | 0.497 | 0.810 | 2.66 | 0.893 | 0.000 | 0.771 | 0.000 | 74.75 | 100.00 | 24.61 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 8 | 44 | 0.407 | 0.552 | 0.815 | 2.63 | 0.888 | 0.000 | 0.741 | 0.000 | 84.48 | 100.00 | 28.41 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 16 | 42 | 0.413 | 0.511 | 0.815 | 2.37 | 0.890 | 0.000 | 0.763 | 0.000 | 80.29 | 100.00 | 24.36 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 16 | 43 | 0.416 | 0.513 | 0.801 | 3.63 | 0.893 | 0.000 | 0.770 | 0.000 | 72.19 | 100.00 | 22.27 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 16 | 44 | 0.413 | 0.491 | 0.799 | 3.28 | 0.892 | 0.000 | 0.759 | 0.000 | 78.41 | 100.00 | 27.83 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 32 | 42 | 0.414 | 0.500 | 0.832 | 2.54 | 0.890 | 0.000 | 0.765 | 0.000 | 81.36 | 100.00 | 26.85 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 32 | 43 | 0.414 | 0.528 | 0.808 | 2.49 | 0.891 | 0.000 | 0.765 | 0.000 | 76.33 | 100.00 | 29.22 | 100.00 | 0.000 | spherocyte stomatocyte |
| MobileNet-UNet | Full (+IPP) | 32 | 44 | 0.601 | 0.595 | 0.831 | 12.47 | 0.928 | 0.000 | 0.770 | 0.707 | 5.96 | 100.00 | 31.92 | 58.25 | 0.000 | spherocyte |